

I N D E X

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S.No.	Date	Title	Page No.	Teacher's Sign/Remarks
		Transportation Engineering I (10 Psn @ 8 marks)		
		Overview		
		AM [1. Introduction (1T)		
		2. Highway Survey (1T)		
		RP [3. Geometric Design (2T+2N)		
		AM [4. Highway Drainage (1T)		
		5. Hill road (1T)		
		PS [6. Highway Materials (LAB) (2T or 1T & 1N)		
		↳ DOR (Manual for test of ME)		
		References		
		• Principles & Practice in Highway Engineering (-LR Kadiyali)		{ [NRS 2070] [ENRS 2071 (2nd rev)] [NRS-2076]
		• Highway Engineering (Khanna & Justo)		
		• Principles of Highway engineering (S.C. Sharma)		
		• Transportation Engineering I (Shrestha & Maxam)		
		• Handouts		
		→ Test on Highway Material		
		→ Standard Specification for road & Bridges (DOR)		
		Assessment day		
		→ submission of Lab report		
		→ Lab report & MCP.		

Chapter 6

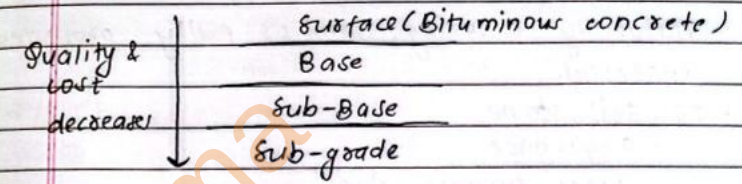
Highway Materials



(i) Flexible Pavement (Bituminous)

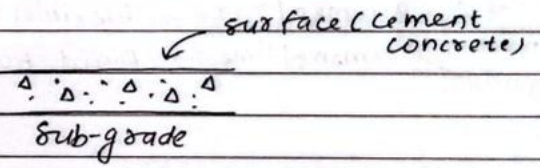
Sub-grade (Natural ground)

↳ Soil of Formation level

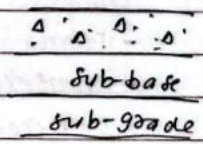


Quality of construction, cost & stress ↓

(ii) Rigid Pavement (Cement concrete)



If weak sub-grade ⇒



6.1 Road construction Material

- Naturally occurring or artificial material used in road construction

Classification:-

(a) Mineral Material

- naturally occurring, semi or fully processed material.
- eg:- soil, stone
- aggregate
- blast furnace slag.

(b) Binding Material

- material mix with mineral to form strong material
- (Mineral binding) • stone dust / clay → semi rigid / semi-flexible bond.
- (Organic binding material) • Bitumen / Tar → Flexible bond.
- (Inorganic) • Cement / lime → Rigid bond

(c) Others

- Paints
- Geotextiles
- Reinforcement
- Gabion wire
- Timber (for formwork)

6.2 Subgrade Soil

- a mineral material formed by disintegration of ~~rock~~ soil
- a supporting soil upon which pavement layers are constructed

6.2.1 Characteristics

(a) Grain size

- ↳ Coarse Grade $> 0.075\text{mm}$
- ↳ Fine Grade $< 0.075\text{mm}$

(b) Shape

- Rounded → alluvial deposits (strong, less-void, less-interlocking)
- Angular → colluvial deposits (Resist deformation, more interlocking)
- Flat & Flaky → weak & not preferred for construction.

(c) Surface Texture (Texture determines water retention capacity)

- ↳ Fine texture - clay / silt (Retain - Not good)
- ↳ Coarse texture - sand / gravel (Permeable)

(d) Chemical composition

Minerals → 45% (Aluminosilicates, oxides, etc)

Organic materials → 5%

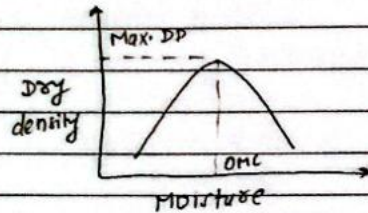
Water - 25%

Air - 25%

(e) Dry density & OMC

→ Soil compacted to Maximum dry density.

- Less voids
- Less permeability
- High strength



6.2.2 Desirable properties

- Stability → Resistance to deformation under vehicular load.
- Incompressibility → don't change in volume upon application of pressure
- Permanency in strength → ability to support pavement at different weather
- Min. change in volume & stability under moisture variation
- Good drainage
- ease in compaction

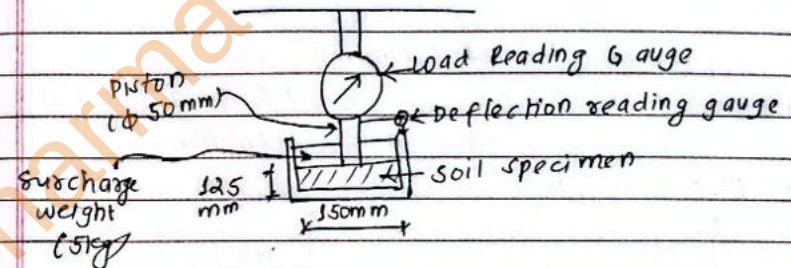
6.2.3 Test

- California Bearing ratio (CBR) Test

$$\rightarrow CBR = \frac{\text{Test Load}}{\text{Standard Load}} \times 100\%$$

→ A relative measure of strength.

→ Developed by California Department of Highway.



→ standard material → Freshly crushed aggregate

② → Loading rate 1.25mm/min

Standard load	Deflection
1370 kg	2.5mm
2055 kg	5.0mm

(for freshly crushed aggregate)

- Specimen preparation

(i) Undisturbed sample → using core cutter

(ii) Disturbed sample

- ↳ Static → Apply load gradually till 125mm high sample obtained
- ↳ dynamic → prepared by compacting in layers.

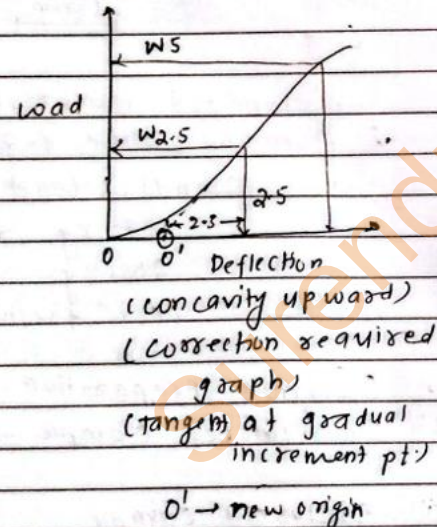
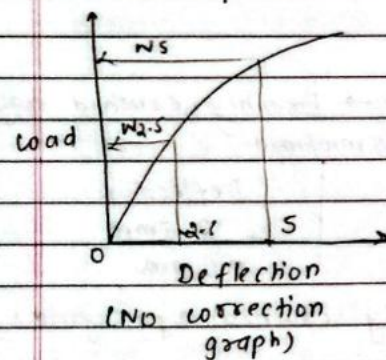
Compaction	Rammer wt.	Fall Ht.	No. of layers	No. of blows
Light	2.5 kg	310 mm	3	56
Heavy	4.89 kg	450 mm	5	56

→ Soil size : 4.75 mm retain
20 mm pass

→ Apply load at 1.25 mm/min

→ Record load for 0, 0.5, 1, 1.5, 2, 2.5, 5, 7.5,
10, 12.5 mm deflection

→ Plot Graph



$$CBR_{2.5} = \frac{W_{2.5}}{1370} \times 100$$

$$CBR_5 = \frac{W_5}{2055} \times 100$$

Generally,

$$CBR_{2.5} > CBR_5$$

If $CBR_{2.5} < CBR_5 \rightarrow$ Repeat the test.

If $CBR_{2.5}$ still $< CBR_5$
after repetition
take CBR_5

For road construction, $CBR > 5\%$.

If $CBR < 5\%$ (for sub-grade)

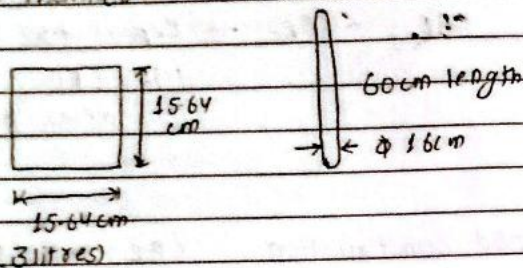
→ Soil stabilization Method

→ Mixing other materials like
cement, lime, etc.

→ Increase embankment height

Angularity Number

The whole number by which the % void exceed 33% when compacted in standard cylinder in specific manner.



Fill cylinder in 3 layers, 100 times tamping

w_1 = wt. of agg + cylinder

w_2 = wt. of empty cylinder

w_3 = wt. of water + cylinder

$$\text{Angularity number (AN)} = 67 - \frac{w_s}{\frac{w_w \times G_s}{\text{solid \%}}} \times 100$$

$$= 67 - \frac{w_1 - w_2}{(w_3 - w_2) \times G_s} \times 100$$

Use,

AN $\Rightarrow$ 0 ~ 11

prefer $\Rightarrow$ 7 ~ 10

AN: 0 11

↑ ↑

Perfectly round aggregate Fairly rounded aggregate

Specific Gravity & water absorption

Bulk specific gravity = $\frac{\text{dry wt. of agg}}{\text{Vol. of agg. including all permeable \& impermeable void}}$

(G_{sb})

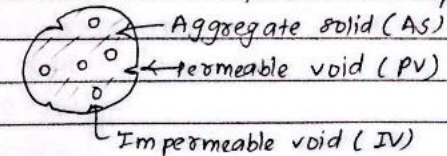
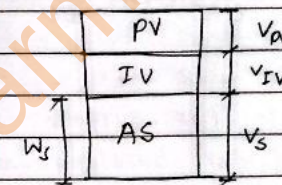


fig:



(Representation by 3-phase diagram)

$$\rightarrow G_{sb} = \frac{W_s}{V_S + V_{IV} + V_{PV}}$$

Practically,

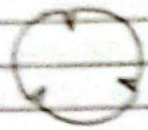
$$G_{sb} = \frac{\text{Wt. of dry Agg. in Air}}{\text{Wt. of agg. in SSD in Air} - \text{Wt. of agg. in water}}$$

Apparent specific Gravity (G_{sa}) = $\frac{\text{dry wt. of Agg.}}{\text{Vol. of Agg. excluding permeable voids}}$

Practically,

$$G_{sa} = \frac{\text{Wt. of dry Agg. in Air}}{\text{Wt. of dry Agg. in Air} - \text{Wt. of Agg. in water}}$$

SSD $\rightarrow$ saturated surface dry



SSD
(saturated surface dry)

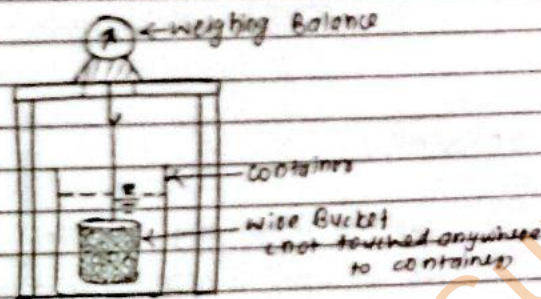
Amount of water absorbed = Wt. of agg. in SSD in Air

- Wt. of dry agg. in Air

% Water absorption = $\frac{\text{Wt. of agg. in SSD in Air} - \text{Wt. of dry agg. in Air}}{\text{Dry wt. of Agg.}} \times 100$

SP. Gr $\rightarrow$ 2.5 - 3 Avg. 2.68

Water Absorption: $< 2\%$ for surface
 $< 4\%$ for bare/rubbase



W_1 = Wt. of agg. + wire bucket in water

W_2 = Wt. of wire bucket in water

Wt. of agg. in water, $W_3 = W_1 - W_2$

Remove aggregate & wipe with cloth

W_3 = Wt. of aggregate in SSD in air

W_4 = Wt. of oven dry Agg. in Air

$$G_{sb} = \frac{W_4}{W_3 - W_2}$$

$$G_{mm} = \frac{W_4}{W_4 - W_{4w}}$$

$$\% \text{ Water absorption} = \frac{W_3 - W_2}{W_4} \times 100$$

Destructive Test

1. Aggregate crushing Value (ACV)

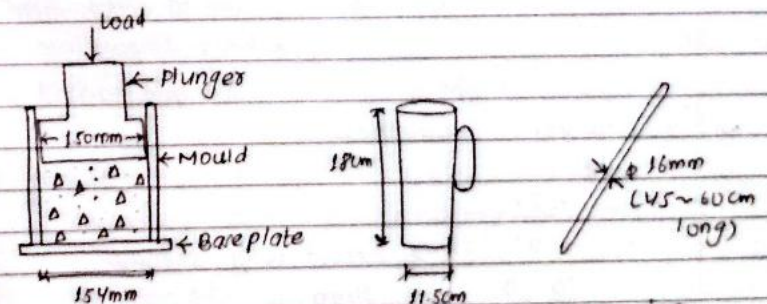
Refers to crushing (gradually applied compressive load)

$$ACV = \frac{\text{Wt. of agg. passing 2.36mm sieve after crushing}}{\text{Original wt. of sample (10-12.5)}} \times 100$$

Use, $\frac{ACV}{\text{Surface}}$

Surface: $< 30\%$

Bare/rub-base: $< 45\%$



weighing balance
Compression Test Machine

(Measuring cylinder)

(ramping rod)

Take aggregate $\rightarrow$ sieve through 12.5mm sieve

Take agg. retained in 10mm sieve

→ Fill in 3 layers, tamping 25 no. of blows

W_1 = Wt. of agg + cylinder

W_2 = Wt. of empty cylinder

W_3 = Wt. of agg. = $W_1 - W_2$

→ Same aggregate fill mould, 3 layers, 25 blows
loading through plunger at 4t per minutes for 10 minutes

→ Crushed aggregate is sieved through 2.36mm

W_4 = Wt. of aggregate passing through 2.36 sieve

Thus

$$ACV = \frac{W_4}{W_3} \times 100$$

2. Aggregate impact value (AIV) / Toughness test

→ Resist to impact load

→ $AIV = \frac{\text{Wt. of agg. passing 2.36mm sieve after crushing impact load}}{\text{original wt. of sample (10-12.5)}} \times 100$

→ measures toughness

→ use:

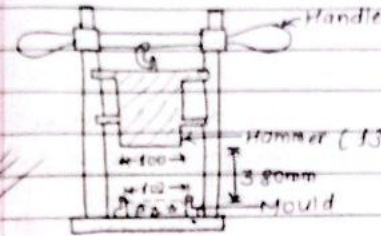
AIV

< 10% → exceptionally strong

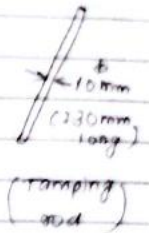
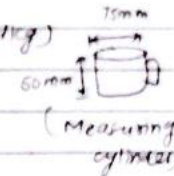
10-20% → strong

20-30% → satisfactory for use in surface

> 35% → Base/sub-base



(Impact test Machine)



→ Weighing balance

→ Impact test machine

→ Take aggregate → 12.5mm pass → 10mm retain

→ 3 layers, 25 nos of light blows

W_1 = Wt. of agg + cylinder

W_2 = Wt. of empty cylinder

W_3 = Wt. of agg = $W_1 - W_2$

→ Fill mould, 3 layers, 25 blows

→ Raise hammer & allows to fall freely for 10 times
(time difference b/w successive dropping not more than 1 second)

→ After test → sieve through 2.36mm

W_4 = Wt. of agg passing 2.36mm

$$AIV = \frac{W_4}{W_3} \times 100$$

(3) Los Angel's Abrasion Test

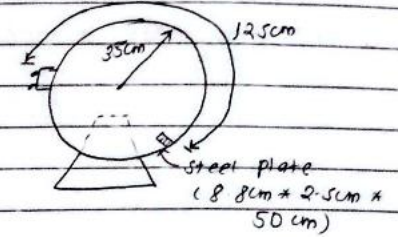
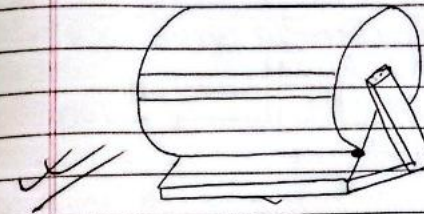
→ resistance to 'wear & tear'

→ measures hardness

→ LA value = $\frac{\text{Wt. of agg. passing through } 1.7\text{mm sieve after abrasion}}{\text{original wt. of aggregate (based on grade)}} \times 100$

Sieve Size		Wt. of sample (kg)							
Passing (mm)	Retained on (mm)	A	B	C	D	E	F	G	
80	63	-	-	-	-	2.5	-	-	
63	50	-	-	-	-	2.5	-	-	
50	40	-	-	-	-	5.0	5.0	-	
40	25	1.25	-	-	-	-	5.0	5.0	
25	20	1.25	-	-	-	-	-	-	
20	12.5	1.25	2.5	-	-	-	-	-	
12.5	10	1.25	2.5	-	-	-	-	-	
10	6.3	-	-	2.5	-	-	-	-	
6.3	4.75	-	-	2.5	-	-	-	-	
4.75	2.36	-	-	-	5.0	-	-	-	

→ A, B, C, D → 5 kg, 500 revolution (33 rev/min)
 E, F, G → 10 kg, 1000 revolution
 Steel balls $\phi 48$, 320g = 442g
 NO. of balls: A, E, F, G → 1 nos
 B → 11 Nos.
 C → 8 Nos.
 D → 5 Nos.



LAA Test

→ Agg. wt. (as per group) & steel balls as per group
 Eg:

Group B: 20-12.5 → 2.5kg } 500 revolutions
 12.5-10 → 2.5kg

NO. of balls = 11

→ After test, sieve from 1.7mm

W_1 = Wt. of original agg.

W_2 = Wt. of agg. retained

→ LA value = $\frac{W_1 - W_2}{W_1} \times 100$

→ Use

Cement concrete < 15%

Bituminous concrete < 30%

Base/sub-base < 50%

Durability Test (Soundness value)

✓ Resistance to weathering

✓ Soundness = $\frac{\text{wt. lost after 10 cycles of heat \& wet in solution}}{\text{Initial wt.}} \times 100$

→ Solvent used: Sodium sulphate or Magnesium sulphate (Na_2SO_4) (MgSO_4)

→ Equipment: wire bucket, sieve, oven, weighing balance

✓ Aggregate?? → Range of agg. size are selected

Retain	Sieve size (mm)	4.75	10	20	40	63	80
✓	Wt. of agg. (kg)	0.3	1.0	1.5	3.0	-	3.0

✓ Test is done separately of each sizes.
(smaller agg. disintegrate fast & large agg. disintegrate slowly)

→ Aggregate in wire bucket

↳ immersed in Na_2SO_4 for 16 hrs

→ Boil 15 min → dry at oven ($105-110^\circ\text{C}$)

↓
Until constant weight is obtained

→ Repeat for 10 cycles

After 10 cycles
• W_1 = Wt. of original aggregate
• W_2 = Wt. of agg. after test
Soundness value = $\frac{W_1 - W_2}{W_1} \times 100$

→ Use, For $\text{Na}_2\text{SO}_4 < 12\%$ } i.e. After 10 cycles, 12% of agg. lost in solvent
For $\text{MgSO}_4 < 18\%$ }
Solvent Soundness value

Bitumen adhesion test:

→ Stripping value Test

→ 5% area not covered by bitumen after aggregate is immersed in water bath of temperature 40°C for 24 hr

→ Static immersion test

✓ 5 agg. → 20 pass 12.5 retain (200 gm)

↳ bitumen → 5% by wt.

✓ heat: agg - 150°C
bitumen - 160°C } separately

→ Add 5% by wt. bitumen in agg. & mix well
(eg. 200 gm agg & 10 gm bitumen)

→ Allow to cool for 1 hr. { room temp }

→ Put mix in container & cover by distilled water by maintaining temperature 40°C .

→ Examine visually and find area not covered by bitumen.

Use
Stripping value $< 25\%$

Gradation Analysis

3 stages:

(i) Gradation test → Sieve Analysis

(ii) Translation / comparing Gradation

PPR (Percent Pass Retain) $\xrightarrow{\text{convert}}$ TPP
(Total Percent passing)

(iii) Combining of aggregate

→ process that determine proportion of aggregate in mix to a final mix is within limit.

Eg. 1

A grading specification is given on the percentage passing-retained basis in Table. Convert these grading requirements to the total percentage-passing by (i) Analytical and (ii) graphical methods.

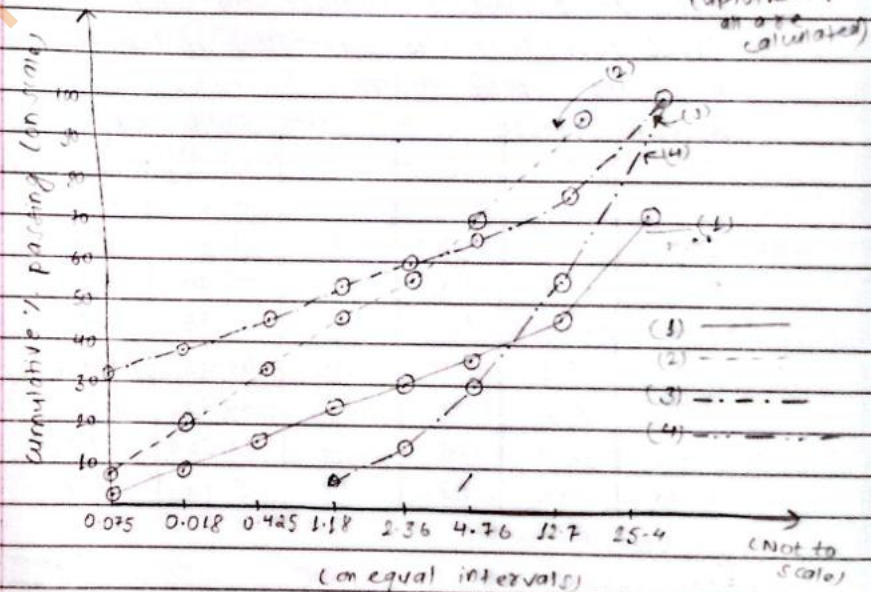
Passing	Retained	Percentage of Material
-	25.4	0
25.4	12.7	25-45
12.7	4.76	10-25
4.76	2.36	6-15
2.36	1.18	6-9
1.18	0.425	8-13
0.425	0.18	7-13
0.18	0.075	7-12
0.075	-	2-8

Converted grading (TPP)

Min → higher of min {max of (1) & (4)}
Max → lower of max {min of (2) & (3)}

Sieve	Cum. % Passing		Cum. % Retained		Cum. % Pass		Cum. % Retain		i.e.
	Minimum	Maximum	Minimum	Maximum	Max.	Min.	Min.	Max.	
25.4	71	140	0	0	100	100	100	100	100
12.7	46	95	25	45	75	55	55	75	55-75
4.76	36	70	35	70	65	30	36	65	36-65
2.36	30	55	41	85	59	15	30	55	30-55
1.18	24	46	47	94	53	6	24	46	24-46
0.425	16	33	55	107	45	-7	16	33	16-33
0.18	9	20	62	120	38	-20	9	20	9-20
0.075	2	8	69	132	31	-32	2	8	2-8
	(1)	(2)			(3)	(4)			

graphically for method (ii) (upto the 1st all are calculated)



(TPP) → for each sieve size

Min → higher of (1) & (4)
Max → lower of (2) & (3)

(iii) Combining of aggregate

→ blending of aggregate

→ dry mix design

→ $P \rightarrow$ % passing as per specification limit for any sieve.

$A, B, C \dots$ i.e. minimum 2 to maximum any no. of agg.

$\hookrightarrow$ % passing of aggregate A, B, C for same sieve

$a, b, c \dots \rightarrow$ proportion of aggregate $A, B, C, \dots$ in the mix to be determined.

$a + b + c = 1$ — (i) (proportion)

$a \cdot A + b \cdot B + c \cdot C = P$ — (ii) (for any sieve)

(w) $a, b, c \rightarrow$ can't be -ve (w) total within specification limit range
i.e. $aA + bB + cC = P$

eg. 2 Mathematical method

Blend an aggregate mixture consisting of three agg. A, B, C such that the gradation of the final mix is within the specified limits.

Sieve size (mm)	Percent passing given			specification limits	
	A	B	C	limits	Mid-point
99.0	100	100	100	100	100
12.5	90	100	100	90-100	95
4.75	40	100	100	60-75	67.5
2.36	6.5	98.1	100	40-55	47.5
0.60	3.0	20.7	93.2	20-35	27.5
0.30	1.2	12.2	58.7	12-22	17.0
0.075	0.5	3.3	27.4	5-10	7.5

↑
(not given in question)

eg

for 2.36 mm,

More eq. for each sieve & solve

$$\begin{cases} a \cdot 6.5 + b \cdot 98.1 + c \cdot 100 = 47.5 \\ a + b + c = 1 \end{cases}$$

$$a + b + c = 1 \text{ — (1)}$$

For 12.5 mm sieve,

$$90a + 100b + 100c = 95 \text{ — (2)}$$

For 4.75 mm sieve,

$$40a + 100b + 100c = 67.5 \text{ — (3)}$$

For 2.36 mm sieve,

$$6.5a + 98.1b + 100c = 47.5 \text{ — (4)}$$

For 0.6 mm sieve,

$$3.0a + 20.7b + 93.2c = 27.5 \text{ — (5)}$$

For 0.3 mm sieve,

$$1.2a + 12.2b + 58.7c = 17.0 \text{ — (6)}$$

For 0.075 mm sieve,

$$0.5a + 3.3b + 27.4c = 7.5 \text{ — (7)}$$

From (2) & (3)

$$50a = 27.5 \Rightarrow a = 0.55$$

Using (1),

$$0.55 + b + c = 1 \Rightarrow (b = 0.45 - c)$$

substituting in (4)

$$6.5 \times 0.55 + 98.1 \times (0.45 - c) + 100c = 47.5$$

$$\Rightarrow c = -0.11$$

rejected (as -ve)

Again,

substituting a, b in (5)

$$3 \times 0.55 + 20.7 \times (0.45 - c) + 93.2c = 27.5$$

$$\Rightarrow [c = 0.23]$$

(Ave $\rightarrow$ okay)

Thus, $[b = 0.22] \rightarrow$ okay

Again check, $A + B + C = P$ (must be within specification limit)

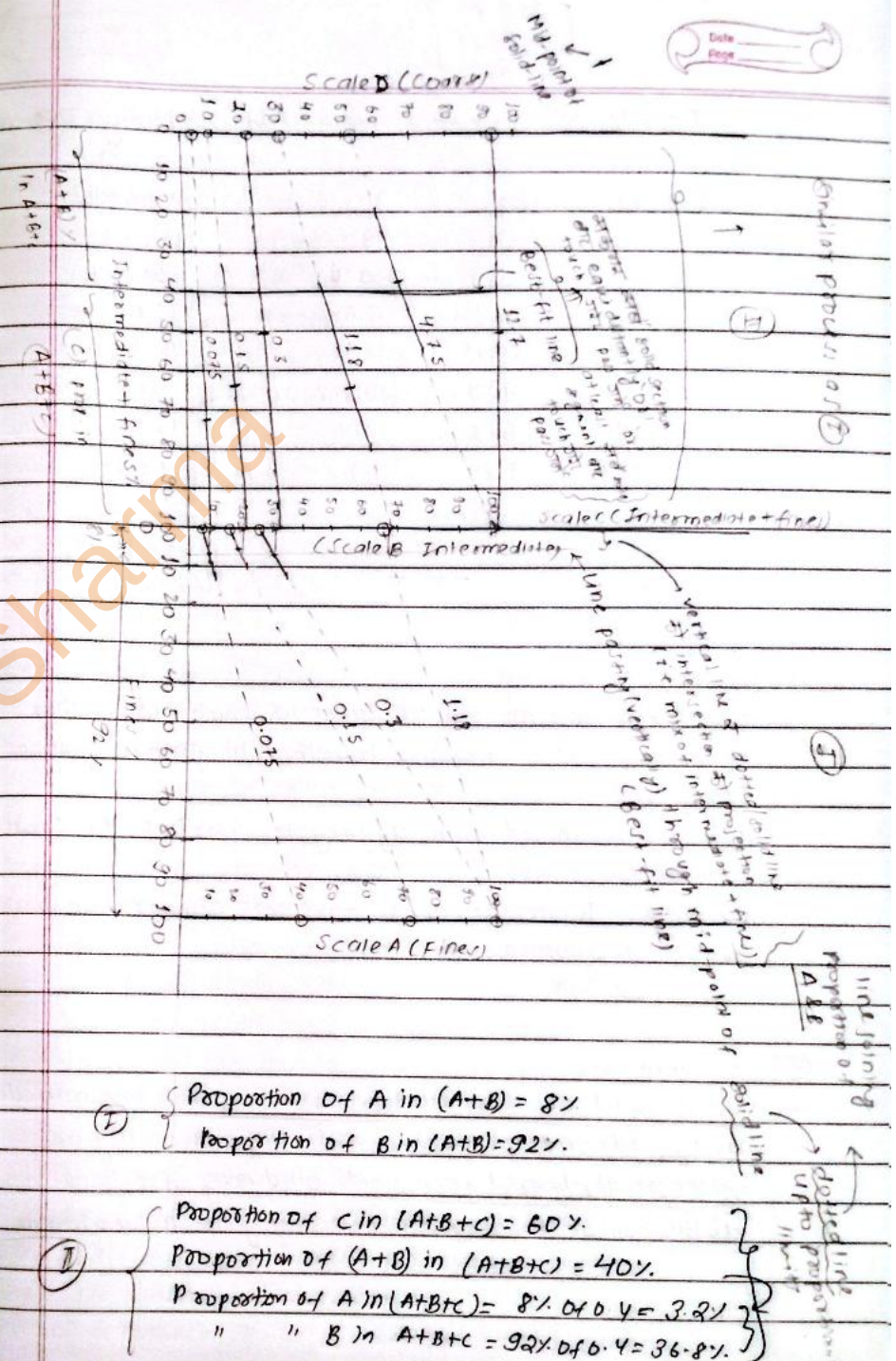
Final Mix				
Sieve Size	Aggregate (A=55%)	Aggregate (B=22%)	Aggregate (C=23%)	Combined
19.0	$100 \times 0.55 = 55$	$100 \times 0.22 = 22$	$100 \times 0.23 = 23$	100
12.5	$90 \times 0.55 = 49.5$	$100 \times 0.22 = 22$	$100 \times 0.23 = 23$	94.5
4.75	$40 \times 0.55 = 22$	$100 \times 0.22 = 22$	$100 \times 0.23 = 23$	67.0
2.36	$6.5 \times 0.55 = 3.6$	$98.1 \times 0.22 = 21.6$	$100 \times 0.23 = 23$	48.2
0.60	$3 \times 0.55 = 1.65$	$20.7 \times 0.22 = 4.5$	$93.2 \times 0.23 = 21.4$	27.6
0.30	$1.2 \times 0.55 = 0.66$	$12.2 \times 0.22 = 2.68$	$58.7 \times 0.23 = 13.5$	16.9
0.075	$0.5 \times 0.55 = 0.275$	$3.3 \times 0.22 = 0.726$	$27.4 \times 0.23 = 6.302$	7.3

Okay.

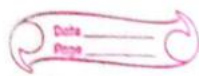
Eg-3 Graphical Method

(9) Blend an aggregate mixture consisting of three Aggregate A, B, C such that the gradation of the final mix is within the specified limits

BS Sieve Size (mm)	Per-cent passing given sieve size				Specifications	
	Fine Agg. (A)	Intermediate Agg. (B)	Coarse Aggregate (C)		Limits	Mid-point
25.4	100	100	100		100	100
12.70	100	100	94.0		90-100	95
4.76	100	100	54		60-75	67.5
1.18	100	66.4	31.3		40-55	47.5
0.30	100	26.0	22.8		20-35	27.5
0.15	73.6	17.6	9.0		12-22	17
0.075	40.1	5.0	3.1		5-10	7.5



$$\begin{cases} a = 3.2\% \\ b = 36.8\% \\ c = 60\% \end{cases}$$



Result of Graphical Method of combining aggregate

Sieve	3.2%	36.8%	60%	Combination
25.4	3.2	36.8	60	100
12.7	3.2	36.8	50.564	96.4
4.75	3.2	36.8	32.4	72.4
1.18	3.2	24.43	18.78	46.012
0.3	3.2	9.57	13.68	25.56
0.15	3.2	6.48	5.4	13.56
0.075	3.2	1.84	1.8	4.506

अगर range fix है तो
Graph exact data में
change a, b, c to make
all in range

* Binder

- ✓ A binder is any material or substance which holds other materials (road aggregates) together to form a stronger component.
- ✓ A binder is mixed with aggregate particles to create bituminous concrete.
- Bituminous binder used in pavement construction works are
 - Bitumen
 - Tar

(a) Bitumen

- viscous liquid or solid material black or dark brown in color, having adhesive properties consisting essentially of hydrocarbons, derived from crude oil during fractional distillation or from asphalt occurring in natural form.

- It is soluble in carbon di-sulphide & in carbon tetrachloride.
- Bituminous materials are very commonly used in highway construction because of their binding & water proofing properties as well as because of relatively low cost. The different grades of bitumen used for pavement construction work of roads and airfields are called paving grade bitumen and those used for water proofing of structures and industrial floors etc are called industrial grade bitumen.
- Are called hydrocarbon binder.
- obtained by distillation of petroleum crude.

(b) Tar

- ✓ - Viscous black/brown liquid obtained as byproduct of destructive distillation of organic material such as coal or wood.
- byproduct of coke industry for steel plant or gas industry.
- Tar is no longer used for paving applications because of its undesirable characteristics including high temperature susceptibility and harmful effects of its fumes during heating.
- Tar proven to cause cancer from occupational exposure & hence its use has been reduced/stopped.
- Two types: coal tar & wood tar
- Types of Tar
 - RT1: very low viscosity, used for surface painting at cold
 - RT2: standard surface painting
 - RT3: surface painting & premixing chips
 - RT4: premixing macadam
 - RT5: highest viscosity, used for grouting purpose.



Bitumen Emulsion

✓ Bitumen (55-75%), water (42-22%), emulsifying agent (3%)

✓ Advantages:

- no heating
- no pollution } compared to cut-back.
- safe to use
- can be applied in moist environment

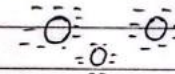
✓ Disadvantages

- emulsifying agent ^{effect} doesn't last long.

• Types (Based on charge)

(i) Anionic

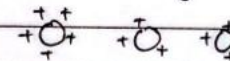
- electro-negative charge
- ✓ - fatty acid and lignin mixed with NaOH.



- useful to apply on aggregate with +ve charge
eg. limestone

(ii) Cationic

- electro +ve charge
- ✓ - amines and animal fats. } emulsifying agents.
- useful to apply on aggregate with
-ve charge e.g. sandstone.



• Based on setting time

(i) RS (Rapid setting)

- Breaks quickly in contact with aggregate
- use for surface dressing
- sets quickly (< 15 minutes)

(ii) Medium setting (MS)

- semi labile
- doesn't break quickly. (More than 1 hr.)
- use in open graded, cold mixer

(iii) Slow setting (SS)

- stable
- need much longer time (≈ 24 hrs)
- use in prime coat, slurry seal
(seal pavement cracks)

□ Cut Back

- viscosity of the bitumen is reduced by using solvent

Advantages:

- no heating
- can be stored & transport for long duration

Disadvantages

- pollution
- safety (as petrol/diesel is highly inflammable)
- solvent are costly.

• Based on curing-setting

(i) Rapid curing (RC)

- highly volatile solvent (naphtha, petrol)
- surface dressing

(ii) Medium curing (MC)

- Mixed with kerosene
- e.g. dense graded, tack coat

(iii) Slow curing (SC)

- Mixed with diesel
 - e.g. crack sealing, prime coat
- (उपरी layer डालि स्केपवि अको layer फाटु अन्दा अगति bitumen layer spread गरिन्छ) (only in impervious layer) (only on previous layer)

Tests

• consistency Test

- measure flow properties of bitumen.

(1) Penetration value Test

Penetration value: the distance in 10th of mm standard needle penetrate into bitumen when standard load (100 gm) is applied for 5 sec at standard temp^o (25°C)

eg. 8.5 mm → 85
(Penetration) (Penetration value)

Penetration grade (PG) 80/100
र जुन नखिन्छ छ लाग्छि

PG grade: 30/40, 40/50, 60/70, 80/100, 175/225
↑ Hot climate ↑ Cold climate

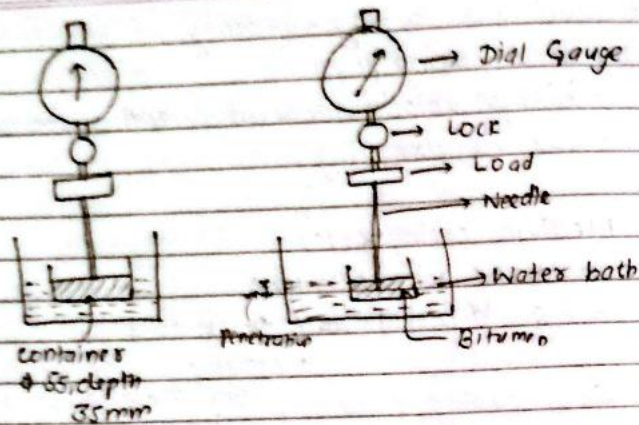


fig: Penetrometer

- Heat bitumen at $75-100^{\circ}\text{C}$ then, pour into container & cool at room temperature
- Keep container in water bath (25°C) → Generally for 30 minutes
- Take out container and put it in penetrometer such that lower needle just to touch surface of bitumen.
- unlock the lock & start the stopwatch simultaneously
- Then, lock after 5 sec and note penetration value

(ii) Viscosity test

- nearest 0.1
- ↳ time in second required to flow 50ml of bitumen through standard orifice at standard temperature.
 - ↳ orifice → 10mm ϕ → bitumen → 40°C
 - ↳ 4mm ϕ → cutback emulsion → 25°C

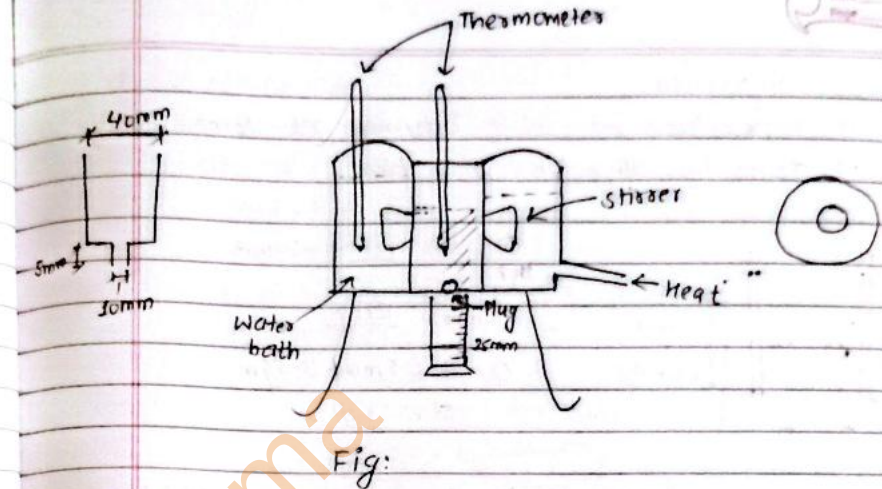


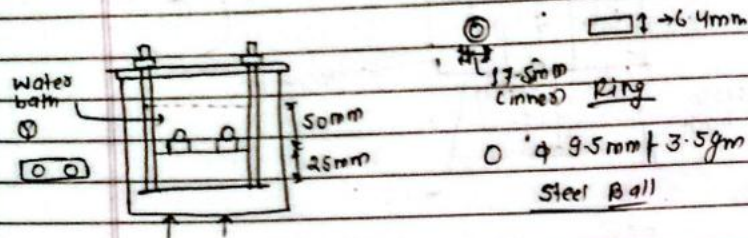
Fig:

- Heat → Pour into tar cup and allow to cool upto $2-3^{\circ}\text{C}$ above test temperature
- Prepare water bath & fit cup into viscometer
- Take 25ml kerosene in cylinder and place below viscometer
- Unplug and start stopwatch
- ↳ Bitumen flow through orifice and collected in measuring cylinder below
- Stop stopwatch when 50ml bitumen collected
- Note time

More time → More viscosity → used in hot climate.

(iii) Softening point

- Temperature at which bitumen get some fluidity
- done by ring and ball test.

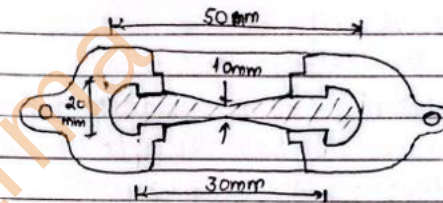


→ Softening point is the temperature at which bitumen can't support wt. of steel ball & start flowing

- Heat → Pour into ring & cool
- Fit ring into frame & put steel ball above using centering ring.
- Frame → immerse in water bath
- Heat @ $5^\circ\text{C}/\text{min}$
- Bitumen soften → start flow with ball
- Note temperature at which bitumen touch lower plate.
- Soften at $35-70^\circ\text{C}$
- Higher softening point → Less temperature stability
(less effect of temperature)
- use Hot climate

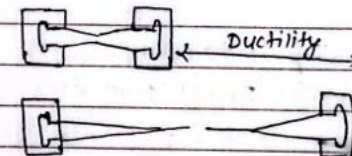
(iv) Ductility Test

- distance in cm, the standard briquette of bitumen will stretch without breaking at standard temperature (27°C)
- Pull rate: 1.25 cm/min
- Higher ductility → Good cohesion & Adhesion
- Minimum 50cm



8-shaped briquette.

- Heat → Pour into standard briquette and cool
- Place in water bath at 27°C for 30min & cut excess bitumen.
- Fit briquette into ductility machine & pull @ 1.25 cm/min .



- Note distance when b/w thread breaks.

Consistency Test

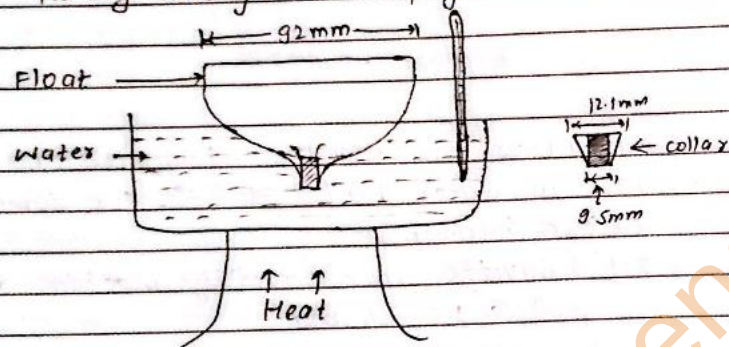
- Penetration test
- Viscosity test
- Softening point test
- Ductility test

Lab

Float test

- Use when viscosity $> 140 \text{ sec}$
penetration < 30

- the time taken from the instant float is placed on water bath (50°C) to the water breaks its way through bitumen plug



- Heat bitumen $\rightarrow$ pour to collar & cool
- place on ice water (5 min) trim excess bitumen
- fit collar into float & put into ice water
- Prepare water bath (50°C) & float the float & start time.
- Note time when water enters float

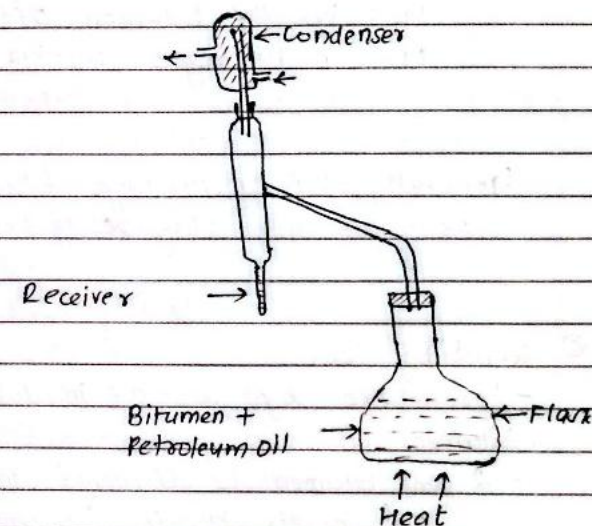
Composition Test

- determine water content, volatile material, insoluble material damaged bitumen.

(a) Water content test

$$\% \text{ water content} = \frac{\text{Wt. of water}}{\text{Wt. of bitumen sample}} \times 100$$

- Distillation test $\rightarrow$ Dean & Stark apparatus
- Max. value = 0.2%
- 100 gm bitumen dissolve in 100 ml petroleum oil
- Dean & Stark method (Distillation test) is used.



- fit distillation apparatus
- Heat $\rightarrow$ water evaporate & condense collected in receiver
- continue till water cease to appear
- Collected water is expressed in terms of mass percentage of sample

Loss on Heating Test

- Determine presence of volatile element
- Loss on Heating = $\frac{\text{Wt. of bitumen} - \text{Wt. of bitumen after heating}}{\text{Wt. of bitumen}} \times 100$

(volatile element escape out on heating)

- Bitumen sample is taken in container
(Wt. of bitumen = (Wt. of bitumen + container) - Wt. of container)

- Heat at 163°C for 5 hr in oven

- Cool $\rightarrow$ Weigh

W_1 = Wt. of original bitumen

W_2 = Wt. of bitumen after heating

Loss on heating = $\frac{W_1 - W_2}{W_1} \times 100$

Generally, - loss on heating $< 1\%$

~~3~~ - Loss on heating $> 2\%$ for penetration value > 150

© Solubility test

- Percentage by wt. of insoluble material present in bitumen
- 2 gm bitumen is dissolved in solvent (carbon disulphide, carbon-tetrachloride, benzene)
- Filtered through Gooch crucible



- Residue washed, dried & weighed

W_1 = Wt. of bitumen

W_2 = Wt. of residue

Insoluble material = $\frac{W_2}{W_1} \times 100$

Insoluble material $< 1\%$

(d) Spot (cracking) test

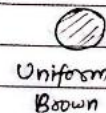
- determine whether bitumen is damaged during heating (in different processing time bitumen is heated)
- +ve result $\rightarrow$ cracked bitumen (Bitumen damaged due to overheating i.e. carbon formed)

How?

- 2 gm bitumen dissolved in 50 ml naphtha
- A drop of bitumen solⁿ is dropped on filter paper.

basically spot on filter paper

- Uniform brown = Negative result
- Brown with black centre = positive result



Uniform Brown



Brown with Black Centre

+ve Result $\rightarrow$ Bitumen not appropriate to use.

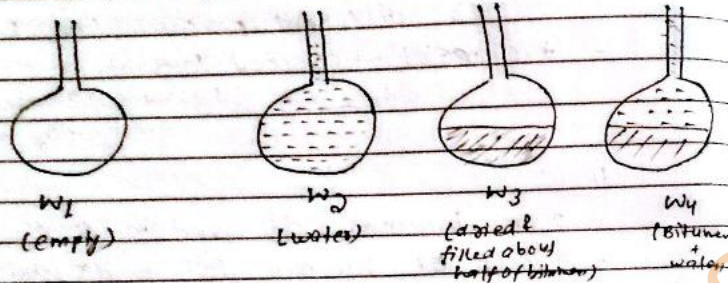
Specific Gravity Test

$$\text{Sp. Gravity} = \frac{\text{Wt. of bitumen}}{\text{Wt. of equal vol. of water}} \times 100$$

- Range: 0.95 - 1.05
- Test temperature: 25°C

How?

W_1 = Wt. of empty bottle (Pycnometer)



W_2 = Wt. of bottle + water

W_3 = Wt. of bottle + Bitumen

W_4 = Wt. of bottle + bitumen + water

$$\text{Specific Gravity} = \frac{(W_3 - W_1)}{(W_2 - W_1) - (W_4 - W_3)} = \frac{\text{Bitumen only}}{\text{only water} - \text{(water above bitumen)}}$$

Safety Test

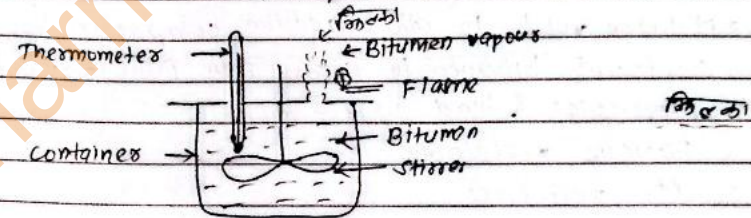
- Flash & Fire point test

- Flash point: Lowest temperature at which bitumen vapour catch fire in the form of flame but fire doesn't last longer.

- Min^m flash point: 175°C

- Fire point: lowest temperature at which bitumen vapour catches fire & burns for at least 5 sec

- Min^m fire point: (175 + 5)°C



How?

- Take bitumen in container & heat

- Start test from the temperature 17°C below expected flash point (@ 1°C rise)

- Note temp. when vapour catch fire in the form of the flame.

- Continue heat & test flame @ 2°C rise

- Note temperature at fire point when fire last at least 5 seconds.

(Difference b/w flash point & fire point = 5°C)

Eg: Flash-180
Fire-182 } Not possible safety issue
Flash-175 }
Fire-175 }

Bituminous mixes

→ Consists of

- coarse agg. - Provide strength
- Fine agg. - Fill voids & make denser structure $\Rightarrow$ more strength
- Filler - ($< 75 \mu m$) $\rightarrow$ make bitumen stiff
- Bitumen - Bind above components

• Desirable properties of Bituminous mixes:-

- Sufficient stability to resist displacement
- Sufficient flexibility under load w/o cracking
- Minimum void to allow additional compaction by Traffic
- Sufficient bitumen to form thin film around aggregates & bind aggregate.
- Fatigue resistance
- Skid resistance
- Workability.

• Types of B. Mixes

(i) As per gradation

- Well-graded (dense)
- Gap-graded (Good tensile fatigue strength, Impermeable)
- Open-graded (Permeable but good resistance)

(ii) As per temperature of Mixes.

- Hot-mix
- Cold mix

(iii) As per site of mix

- Plant Mix (Quality control)
- Road Mix

Design of Bituminous Mix

Selection of Aggregate

As per gradation
 - Flatness & elongation Index
 - Different values as per specification (LA value, Impact value, etc.)

Selection of Bitumen

As per temperature

Optimum Binder

Surface-Area Concept

Void concept
 (Marshall Mix design)

(1) Surface area concept

- determine amount of binder just enough to cover the surface of all agg.
- low road, initial approximation
- Nebrascan Formula

$$P = AG(0.02a) + 0.06b + 0.1c + Sd$$

P = % of binder

A = absorption modifying factor = 1 for Sand & Gravel
 = $1 + 0.67 \times$ Water absorption % for others

G = Sp. gr. correction factor for agg. passing 300 μ m = 2.62
 Ap Sp Gr

a = % by wt. of agg. retain 300 μ m

b = % " " " " pass 300 μ m retain 150 μ m

c = % " " " " " 150 μ m " 100 μ m

d = % " " " " " 100 μ m " 75 μ m

S = experimental factor (0.2)

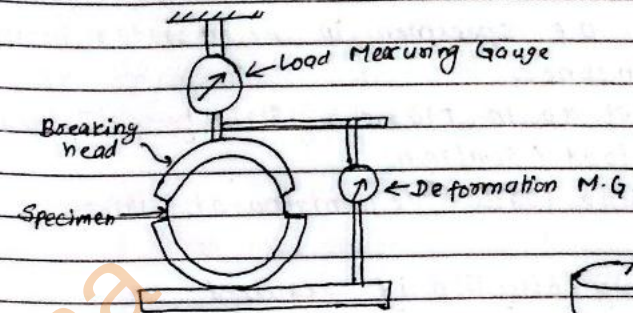
(2) void concept (Marshall Mix design)

2 Stage

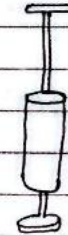
i. Marshall Stability test

ii. Density Void Analysis.

i. Marshall Stability Test



Mould - 100 mm ϕ (101.6 mm)
 Rammer: 63.5 mm height with collar



(3) Specimen preparation

≈ 1.2 kg

- heat agg $\rightarrow 154 \sim 160^\circ\text{C}$

- " bitumen $\rightarrow 120 \sim 125^\circ\text{C}$

- add total bitumen content in agg. & mix.

- all content $\rightarrow$ mould $\rightarrow$ compaction $\rightarrow$ 50 nos $\rightarrow$ both sides
 cool

$\rightarrow$ remove with help of extender.

(drop ht = 457 mm) - 75 blows on either side

II Test

- immerse water bath (60°C) for 30 min
- wt. of specimen in air, in water, in air in SSD, thickness
- set up in Marshall stability equipment & apply load (5cm/min)
- note load & deformation at failure

III Apply correction if required

If thickness $\neq 63.5\text{mm}$

Thickness	Corr ⁿ
57.1mm	$\rightarrow 1.19$
63.5mm	$\rightarrow 1.0$
69.9mm	$\rightarrow 0.86$

$$\text{Corrected stability} = \text{cor. factor} \times \text{Stability}$$

IV Optimum binder

$$\text{OBC} = \frac{B_1 + B_2 + B_3}{3}$$

B_1 = Bitumen content for max. density
 B_2 = " " for max. stability
 B_3 = " " for 4% air void

(ii) Density Void Analysis

- # Determine optimum bitumen content. Also determine effective sp gr of mix, Absorbed bitumen, effective bitumen.

$$\text{Bulk sp.gr. of agg } (G_{sb}) = 2.63$$

$$\text{Sp.gr. of bitumen } (G_b) = 1.02$$

% bitumen	Max. Theor.	Wt. of specimen			thickness (mm)	Stability (kg)	Flow (mm)
	Sp Gr. of mix (G_{mm})	in air	In water	in air in SSD			
4.5	2.47	1240	713	1248	66.2	1043	3.3
4.75	2.46	1244	723	1252	65.4	1170	3.6
5.0	2.45	1239	723	1245	64.5	1228	4.2
5.25	2.44	1250	731	1256	64.9	1134	4.4
5.5	2.43	1248	729	1254	64.8	992	4.7

$$\text{Theoretical sp gr } G_{mm} = \frac{W_a}{V_c + V_f + V_{mf} + V_b}$$

$$W_a = \text{Wt. of mix in air}$$

c $\rightarrow$ coarse agg.

f $\rightarrow$ fine agg

mf $\rightarrow$ mineral filler

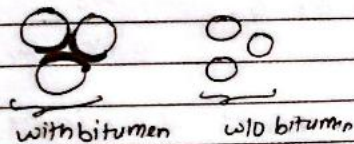
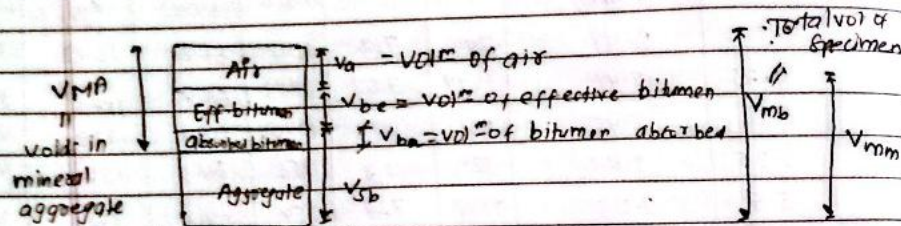
b $\rightarrow$ bitumen

$$G_{mm} = \frac{W_a}{\frac{W_c}{G_c} + \frac{W_f}{G_f} + \frac{W_{mf}}{G_{mf}} + \frac{W_b}{G_b}}$$

(i) Bulk specific gravity of specimen

$$G_{mb} = \frac{\text{Wt. of specimen in air}}{\left(\frac{\text{Wt. of specimen in air}}{G_{mb}} \right) - \left(\frac{\text{Wt. of specimen in water}}{G_{mb}} \right)}$$

$$\text{For 95\%} \quad G_{mb} = \frac{1240}{1248 - 713} = 2.32$$



(ii) % air void

$$P_a = \frac{V_a}{V_{mb}} \times 100 = 100 - \frac{100 \times G_{mb}}{G_{mm}}$$

$G_{mb} \rightarrow$ Bulk sp. gravity

$G_{mm} \rightarrow$ Theoretical sp. gravity

For 95%,

$$\text{For 95\%} \quad P_a = 100 - \frac{100 \times 2.32}{2.47}$$

$$= 6.07\%$$

(iii) Voids in Mineral Aggregate (VMA)

$\rightarrow$ Space between aggregate particles (after removing bitumen)

$$VMA = 100 - \frac{G_{mb}}{G_{sb}} \times P_s \quad \left\{ \begin{array}{l} \text{For 95\%} \\ = 100 - \frac{2.32}{2.63} \times 95.5 \\ = 15.76\% \end{array} \right.$$

$P_s \rightarrow$ % aggregate

For 95%,

$$P_b = 4.5 \text{ (\% bitumen)}$$

$$\% \text{ of agg, } P_s = 100 - 4.5 = 95.5\%$$

(iv) Voids filled by bitumen (VFB)

$$VFB = \frac{VMA - P_b}{VMA} \times 100$$

For 95%,

$$VFB = \frac{15.76 - 6.07}{15.76} \times 100 = 61.48\%$$

P_b	P_s	G_{mb}	P_a	VMA	VFB	Corrected stability	Flow
4.5	95.5	2.32	6.07	15.76	61.48	980.4	3.3
4.75							
5.0							
5.25							
5.5							

(v) Correction factor,

For, thickness = 66.2 mm

$$\begin{array}{l} 57.1 \rightarrow 1.19 \\ 63.5 \rightarrow 1.0 \\ 69.9 \rightarrow 0.86 \end{array} \left. \begin{array}{l} 66.2 \rightarrow \\ 66.2 \rightarrow 0.94 \end{array} \right\}$$

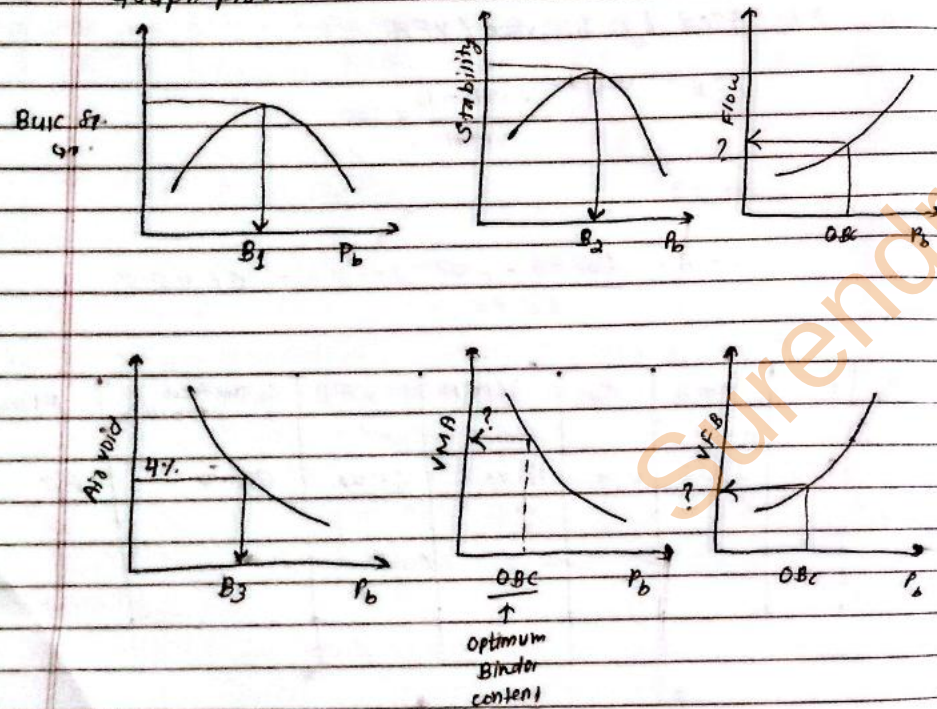
$$\text{Stability} = 1043 \text{ kg}$$

$$\text{Correction factor} = 0.94$$

$$\begin{aligned} \text{Corrected Stability} &= 0.94 \times 1043 \\ &= 980.4 \text{ kg} \end{aligned}$$

Do same for all & fill table.

Graph plot:



(?) $\rightarrow$ check with standard.

Additional,

$$\text{Effective sp. gr. of Mix} = G_{se} = \frac{P_s}{\frac{100}{G_{mm}} - \frac{P_b}{G_b}}$$

$$\text{Absorbed bitumen} = P_{ba} = G_b \left[\frac{G_{se} - G_{sb}}{G_{se} + G_{sb}} \right] \times 100$$

$$\text{Effective bitumen} = P_{be} = P_b - \frac{P_{ba} \times P_s}{100}$$

Bitumen Extraction Test

- to determine actual bitumen content in mix
- $\% \text{ bitumen} = \frac{\text{Wt. of mix} - \text{Wt. of agg (dry) after extraction test}}{\text{Wt. of mix}} \times 100$

- Centrifuge extractor
- Solvent - benzene, carbon tetrachloride,
- About 1-2kg of mix in bowl & covered with solvent
- operate 3600 rpm
- when liquid cease to come out add more solvent
- continue till clear liquid
- remaining agg., dried & weigh

- A hill road is usually defined on the basis of terrain.
- As per NPS 2070 the cross slope may be classified as:

6-1. Terrain classification:

S.NO.	Terrain Type	Percent cross slope	Degree
1	Plain	0-10	0°-5.7°
2	Rolling	>10-25	>5.7°-14°
3	Mountainous	>25-60	>14°-31°
4	Steep	>60	>31°

- Hill road - one which passes through terrain with cross-slope of 25% or more i.e. mountainous or steep
- There are sections along hill roads with cross slope less than 25%, esp. when the road follows river route.

- Importance of Hill road
 - Economic development of country
 - For industrial development
 - Development of Tourism
 - To establish other projects such as Hill irrigation, Hydropower, etc.
 - To meet the need of strategic consideration of country in hilly area of country (Strategic Road Network - SRN)
 - In mountainous and steep terrain, navigation and rail facilities are not possible which automatically demands hill road.

✓ Why hill road?

- 66% covered with thinly / sparsely populated hills, mountains.
- Gov. population exists in hilly areas and depends on agro products
- Waterways, railways, airways → Even not possible or very difficult

[Railway → req^d cross slope : 1-2%]

[Airports → long straight runway required]

- Hilly regions → Extremes of climatic conditions; Difficult and Hazardous Terrain Topography and vast High altitude areas.

[Fragile lands, steep terrain, landslides, snowfalls → effective wt ↑ → landslides ↑
Freezing and thawing of pavement
Moisture variation in subgrade]

- Challenger of Hilly areas: Floods, Landslides, snowfalls, etc.
↳ Fairly weathered Roads possible

- But Hilly areas Rich in Natural resources, flora, fauna
Important to launch development projects
(Eg: Hydropower, etc), tourism, etc
National Strategic consideration.

- For Transport, Economy, social reasons → Justify the construction of Hill roads in the hilly regions
we see social aspects more rather than economic (B/C, NPV).
- Where Tunnel, viaduct → possible but not economical

• Design and construction problems

- Highly broken relief with widely differing elevations, steep slope, deep gorges, and great number of water courses. This may unnecessarily increase road length.
- Slope unstable after road construction
- Great variation in hydrological condition (Rainfall, snowfall)
- Great variation in geological condition from place to place. (different soil, rock mass)
- Variation of climatic condition.
- High speed of runoff
- Frequent blasting (Rockfall in Siddhababa, Siddhartha HW).
- Need of special safety precaution. (Sufficient dimension cannot be provided)
- Large number of cross-drainage structures.
- Filling may overload the soft underlying layer may cause landslide
- Complex Geology → Different Rock Bed → Geology differs from place to place → Assessment for road foundation is not easy (GWT, sub-surface drainage)
(Egk. problem unstable tectonic plates)
- Environmental impact → Removal of vegetation, etc. → stable slope may change to unstable after road construction.
(After hill road constⁿ GWT may fall to stream may dry out, then again → vegetation & ecological damage)
- Great variation in hydrological condition → landslides, -
- soil erosion → may lead to damage after road construction.

→ New earth fill for road embankment may overload the relatively weak underlying soil layer on hill slope

↓
may trigger new and recurrent slides.
(filling pattern → proper compaction, soil stabilization, road).

→ Requires installation of various types of special road structures → cost may rise upto 50-60% of total construction

↓
Soil nailing, rock bolting,
shotcreting, lining,
gabion wall, slope protection

→ Steep slopes → high speed of surface runoff → provision of erosion protection works.

(check dams, slope protection work, vegetation cover, wire netting)

→ Construction along relatively small approach stretches
↓ different construction technology may be needed
(diff. soil / rock mass → diff. construction technology)

→ Need of special safety precaution during hill road construction

→ Frequent blasting.

→ Design of hairpin bends to get heights
(for drainage)

SEC → cost for
Safety Economy

Special consideration in Hill Roads

1. Selection of Road Alignment
2. Geometric design (gradient and hair pin bend)
3. Cross-section design
4. Design of drainage structure (special structure may be required)

I # Factors affecting alignment of hill road

1. Temperature.
2. Rainfall
3. Atmospheric pressure.
4. Geological conditions

1. Temperature

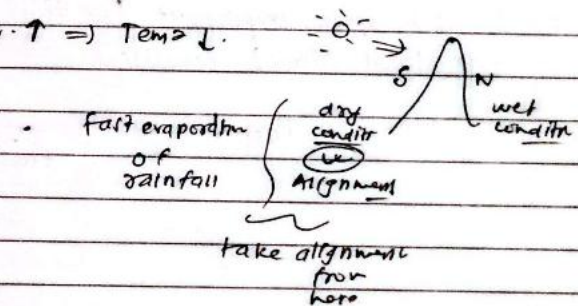
→ Air temp^{re} in hill is lower than in valley

→ Temp^{re} varies inversely with altitude as the temp^{re} drop being about 0.65°C per 100m rise.

→ Amt. of sunlight depends on orientation of hill slope

→ Slope facing south or southwest receive more sunlight hence snow disappears quickly and rain water evaporates quickly.

Elev. ↑ ⇒ Temp ↓.



2. Rainfall

→ As the elevation increases, rainfall also increases upto certain height.

→ Maximum rainfall occurs at zone of intensive cloud formation (1500m to 2500m) above MSL.

→ 40-60mm rainfall increase in rainfall for every 1000m of elevation.

→ In summer, very heavy storm occurs and about 15-20% of annual rainfall may occur in one storm. (P&T notes)

→ At many locations in hill roads, the landslide and slip occur locking traffic during heavy rainfall.

→ Large number of cross-drainage structure may required.

Monsoon - 90% of annual rainfall.

3. Atmospheric pressure

→ Atm. press. decreases with increase in elevations

→ At altitudes (8000-4000m), wind velocity is 25-30m/s.

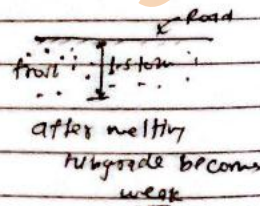
→ Depth of frost penetration 1.5 to 2m (15-20°C)

→ There is intensive rock weathering. (due to alternate freezing & thawing)

→ Low cost roads are damaged by wind. (Low cost road in gravel roads, less than road)

→ So it is very important for the designer to be familiar with local climate condition.

Elev. ↑ ⇒ P_{atm} ↓

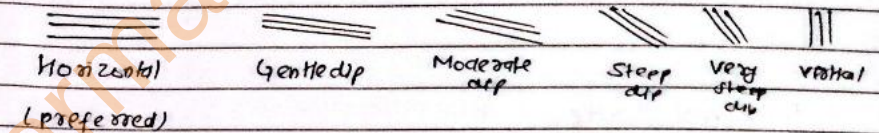


4. Geological conditions

Degree of stability is depended on:

- Type of rock
- Degree of strata inclination (dip)
- Hardness of rock
- Presence of GW

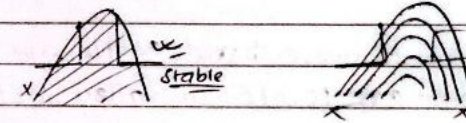
→ Degree of strata should be as small as possible or alternatively be inclined away from the road.



→ Increase in angle of dip.

(Fault, fold, → ignored if possible)

(dip angle minimum → preferred)

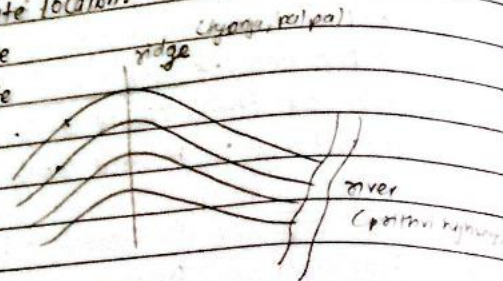


- Route Location (under alignment selection)
↳ difficult task in hill road construction. Hill road is characterized by long route with many sharp curves.

Two possible route locations:

1. River route

2. Ridge route



1. River Route (more preferable)

- Route is located along the river valley.
- Most frequent case of hill road alignment.
- As route is located on valley, comparatively gentle slope.
- It is possible to serve rural settlements, as most of the settlements are located in valley part.
- Low VOC, availability of construction materials (water, sand, aggregate, etc.)
- Large number of horizontal curve, but rarely required hair-pin bend.
- Large number of cross-drainage structures are required.
- Extensive earthwork required.
- Construction of special structure such as retaining structure along hill side.
- Massive river training works are required.

(larger span)

⇒ Special consideration:- (away from sufficient waterway).

- Road bed should be sufficiently above and away from Maximum flood level (MFL).
- If road bed is near watercourse, embankment slope should be adequately stabilized.
- More attention to hydrological and geological structure.
- Select best alternative while crossing the bridge or water courses.

2. Ridge Route (highest pt. of hills)

- It gives very steep gradient.
- Large number of sharp curve including hair pin bend.
- Extensive rock work (hair-pin bend).
- Construction of special structure (Box cut).
- Necessity of long route.
- Geologically stable and mild slope should be selected.

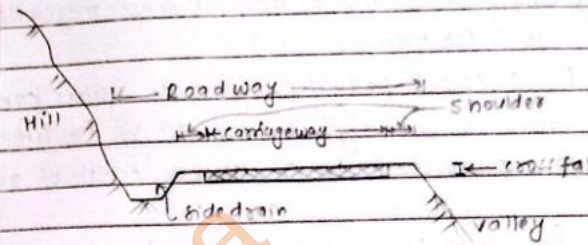
River Route

- Most frequent Gentle slope
- Serves rural settlement situated next to the water course.
- Low vehicle operation cost
- Availability of construction material and water
- Involve numerous horizontal curves, construction of large bridges
- Special retaining protection on hill sides
- River training structures maybe extensive to routing, etc.

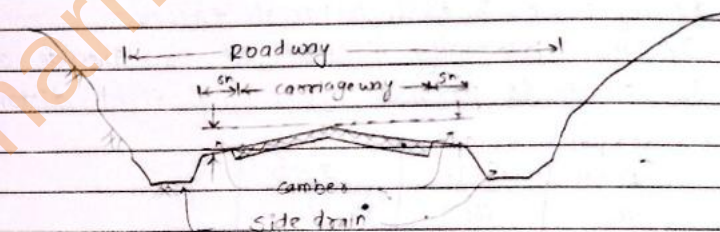
Ridge Route

- Steep gradient sharp curves including hair pin bends
- Extensive rock works
- Successive mountain pass
- Route climbs up continuously from the valley Hill mountain pass and descends down
- Construction of special structures, tunnels, snow fences, etc.
- Are most stable

2. Geometric design of Hill road



(a) Side hill cut



(b) Box cut

- Hill road have to negotiate through difficult topography, inhospitable terrain and extreme climatic conditions.
- Geometric design should be based on traffic safety.
- Development of cross section in stage is technically feasible.
- Improvement of grade and curvature is very expensive and sometimes impossible.

• Sight distance

- On hill roads stopping sight distance is absolute minimum from safety angle and must be ensured regardless of any other considerations.
- It would be a good practice if this value can be exceeded and visibility corresponding to intermediate sight distance provided in as much length of road as possible.
- If this is the case then drivers will be able to get opportunities to overtake with caution and driving task becomes much easier.
- If not normally feasible or practicable on hill roads to provide overtaking sight distance

NRS

(kmph) Speed	(m) SSD	(m) ISD
20	20	40
25	25	50
30	30	60
35	40	80
40	45	90
50	60	120

NRS

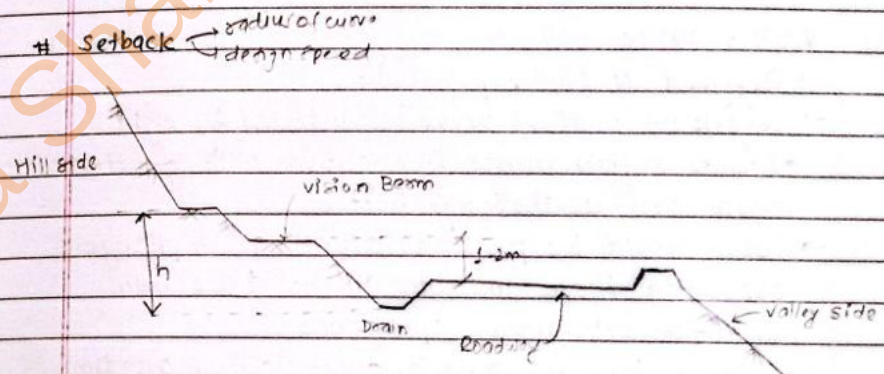
	Driver's eye ht.	Ht. of object
SSD	→ 1.2m	0.15m
ISD	→ 1.2m	1.2m

• Superelevation

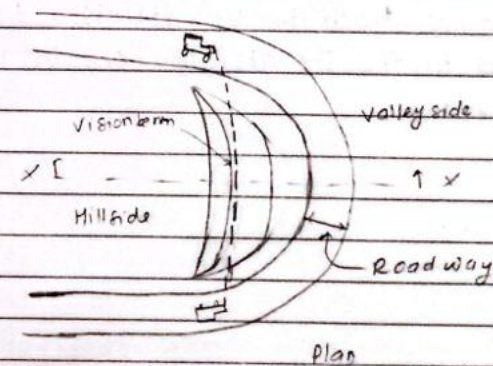
→ Limiting value

- (a) In snow bound areas = 7%
- (b) In hilly areas not bound by snow = 10%

→ In attaining the required superelevation, it should be ensured that the longitudinal slope of the pavement edge compared to the CL (i.e. rate of change of superelevation) is not steeper than 1 in 150 for roads in plain and rolling terrain and 1 in 60 in mountainous and steep terrain.



Section at X-X



Plan

• Gradient

→ A study on dynamic characteristics for the truck indicates that:

- > If gradient is 4% the speed is 42 kmph
- > If gradient is 6% the speed is 32 kmph
- > " " " 8% " " " 24 kmph

> If gradient chosen is 4% instead of 6%, the route will be double in length but the speed will increase only 1.75 times. This indicates in spite of decrease in gradient there is no gain in travel time.

• Vertical curve

→ Designed as square parabola

→ Length of vertical curve is controlled by sight distance requirements but curves with greater length are aesthetically better.

→ Curve should be provided at all grade change exceeding those indicated in the table below and for satisfactory appearance.

→ Where hz. and summit/crest curve overlap, the design should provide for the required sight distance both in vt. direction along the pavement and in the hz. direction on the inside of the curve.

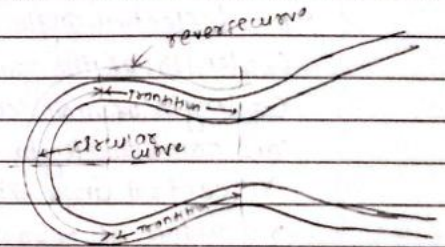
• Passing places

- Required on single lane hill roads to facilitate crossing of vehicles approaching from opposite direction and to pull aside a disabled vehicle so that it does not obstruct traffic.
- Provided at the rate of 2-3 per km.

• Design of Hair Pin Bends

- In hilly regions → difficult to avoid bends → where direction of the road reverse
- In hill road alignment, hair pin bends are introduced particularly → when necessary to attain height w/o covering substantial hz. direction.
- Hair pin bends → designed either as a circular curve with transition at each end or as a compound circular curve.

(Ridge route)

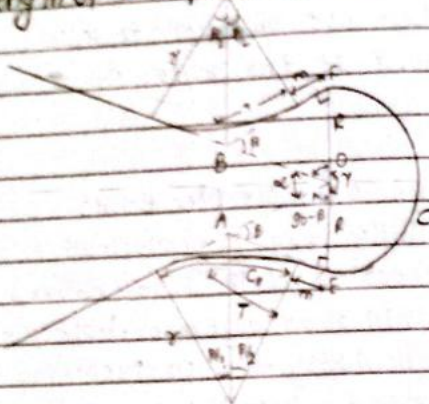


• Design criteria for hair pin bend:

- (a) Straight length b/w two successive hair pin bend should be 60m excluding the length of circular and transition curve.
- (b) Minimum design speed = 20 kmph
- (c) Minimum radius of inner curve = 15m
- (d) Minimum length of transition curve = 15m
- (e) Super elevation in circular portion = ~~1 in~~ 10%
- (f) The Max^m and min^m gradient = 1 in 40 and 1 in 200.
- (g) Approach gradient should not be steeper than 5% for 40m.

- Types:
 - Symmetrical
 - Any minor type

= Expression for length of Hale-pin bend



Here,

T = tangent length of reverse curve

R = Radius of main curve

β = deflection angle

C_2 = length of the reverse curve

C = length of main curve

m = transition length between reverse and circular curve

γ = central angle at centre of main curve (by circular curve)

α = radius of reverse curve.

- Tangent length of reverse curve:

$$T = \alpha \tan \beta/2$$

where,

T → length of the tangent

α → Radius of the reverse curve

β → Deflection angle

T → tangent length of reverse curve

(m → straight line transition)
(min = 15m)

The distance from the apex of the reverse curve angle to the commencement of the main curve is:

$$AE = BF = T + m$$

From $\triangle AOE$ or $\triangle BOF$ it will be found that

$$\tan \beta = \frac{OE}{AE} = \frac{R}{T+m} = \frac{R}{\alpha \tan \beta/2 + m} \quad \text{--- (1)}$$

Where, R is radius of main curve. From trigonometry,

it is also be known that:

$$\tan \beta = \frac{2 \tan \beta/2}{1 - \tan^2 \beta/2} \quad \text{--- (1)}$$

substituting this expression for $\tan \beta$ in eqⁿ (1)

$$\left\{ \tan \beta/2 = \frac{-m + \sqrt{m^2 - R(2\alpha + E)}}{R + 2\alpha} \right\} \quad \text{--- (m)}$$

Here, angle β to correspond to ' R ', ' α ' and ' m ' can be easily determined.

The distance from apex of the reverse curve to the centre of the main curve is determined by:

$$AO = OB = \frac{T+m}{\cos \beta} = \frac{R}{\sin \beta}$$

The central angle γ corresponding to main curve of bend,

$$\gamma = 360^\circ - 2(90^\circ - \beta) - \alpha = 180^\circ + 2\beta - \alpha$$

And the length of main curve is:

$$C = \frac{\pi R \gamma}{180^\circ}$$

Hence total length of bend is

$$[S = 2(C_2 + m) + C] \text{ m}$$

C_2 → length of reverse curve.

For eqn (iii)

$$\frac{R}{\sigma \tan \beta/2 + m} = \frac{2 \tan \beta/2}{1 - \tan^2 \beta/2}$$

$$\text{or, } 2 \tan \beta/2 = \frac{R (1 - \tan^2 \beta/2)}{m + \sigma \tan \beta/2}$$

$$\text{or, } 2 \tan \beta/2 (m + \sigma \tan \beta/2) = R (1 - \tan^2 \beta/2)$$

$$\text{or, } 2m \tan \beta/2 + 2\sigma \tan^2 \beta/2 = R - R \tan^2 \beta/2$$

$$\text{or, } 2\sigma \tan^2 \beta/2 + R \tan^2 \beta/2 - R + 2m \tan \beta/2 = 0$$

$$\text{or, } (2\sigma + R) \tan^2 \beta/2 + 2m \tan \beta/2 - R = 0$$

∴ This is quadratic equation of $\tan \beta/2$,

$$\therefore \tan \beta/2 = \frac{-2m \pm \sqrt{(2m)^2 - 4(2\sigma + R)(-R)}}{2(2\sigma + R)}$$

$$\left[\tan \beta/2 = \frac{-m \pm \sqrt{m^2 + R(2\sigma + R)}}{2\sigma + R} \right] \text{ or}$$

Free haul distance → contractor को extra payment नहीं मिलेगी यदि distance बड़ा हो जाये तो लगाना पड़ेगा

3 Different types of Hill Road cross-sections:

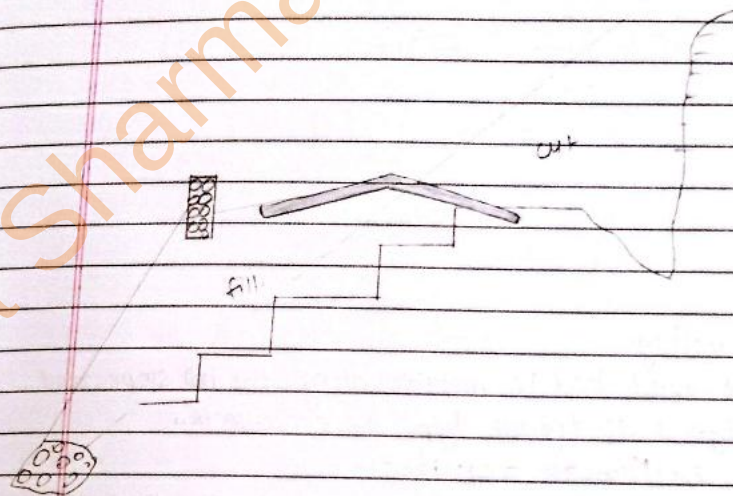
(i) cut & fill

→ If hill side slope is $> 2\%$, cut and fill road is cheaper and environmental friendly.

→ The fill mass is obtained from cut material at some locations or within free haul distance

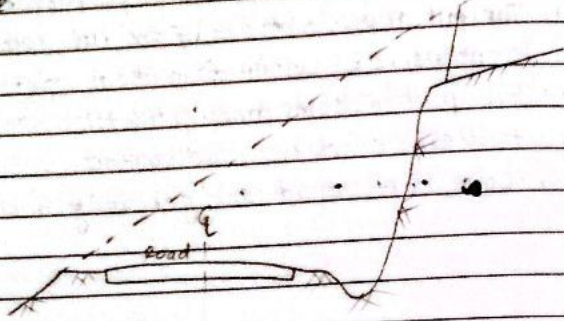
→ Fill portion sometimes get settled and causes longitudinal crack on road surface.

→ Low cost road are generally made with cut & fill.



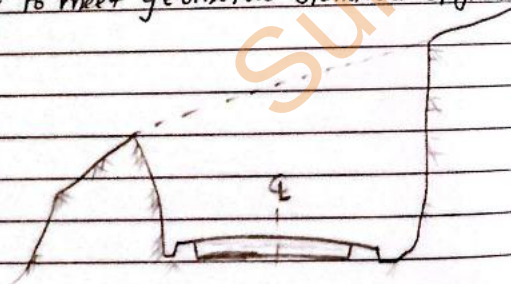
(ii) Bench type full cut

→ with increase in earthwork ensures the complete stability of the hill road bed, if hill side itself is stable.



(iii) Box Cutting

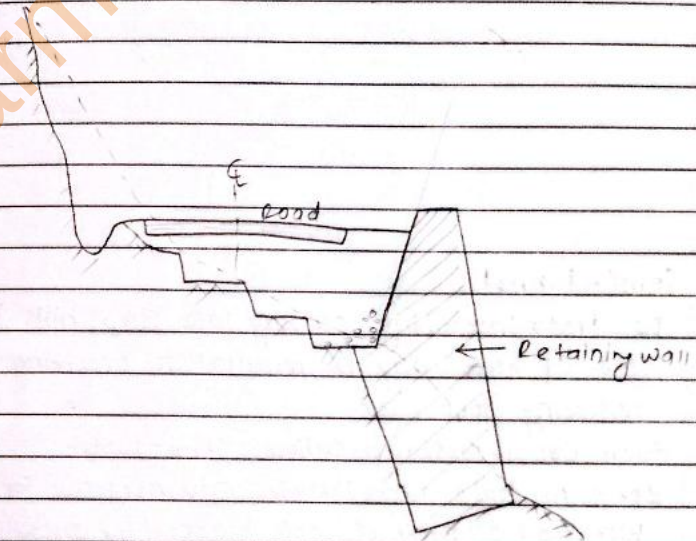
- If the road bed is unstable along the hill slope, road is designed as trench type of cross-section.
- Large earthwork required.
- It is necessary to meet geometric standard of given class of road.



fill material itself is stable (retaining str.)

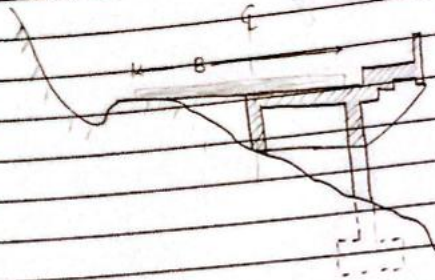
(iv) Embankment with retaining wall

- on steep slope over 30-35 degree, the earthwork involved in constructing embankment increases substantially.
- When natural slope of ground is greater than embankment slope, they never meet downhill.
- Hence, to support the embankment soil, retaining structural walls is necessary.
- Retaining walls are also preferred in less steep slope to increase stability.



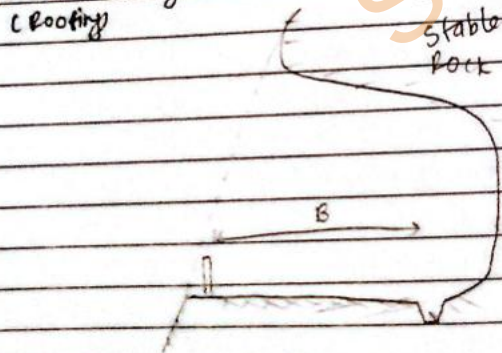
(v) Semi Bridge - if R.W. ht. very large (pm. case) act as formation width

- If the road is located steep slope, the R.W. ht. increases substantially.
- In such cases in order to reduce the quantity of work, semi-bridge type structure may be constructed.
- Part of road is accommodated on bench cut and part is located on the semi-bridge.



(vi) Semi-tunnel

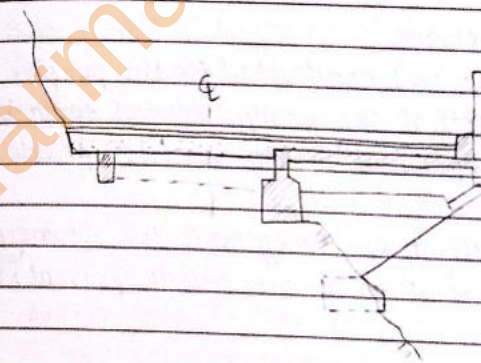
- If ~~the~~ inscribing cutting into steep hills in stable faces of rock may be permitted to overhang the road reducing rock work.
- Such cross-section is called semi-tunnel.
- Road in half tunneling is advantageous for high altitude hill road as they do not blocked by snow.



very steep or vertical cut and earth

(vii) Platform

- In steep slope, where shifting of the route into the hill side will lead to enormous rock work, which increase cost of construction.
- Where semi-tunnel cannot be constructed due to certain geological reason platform are constructed.
- Platform are usually cantilever out of the rock on which roadway is partially located.



4 Special structure in Hill Road
 → Construction of hill roads involve many special structures to retain soil, slope stability and dissipate energy of flowing surface water.

Special structures are:

1. Retaining structure
2. River training "
3. Landslide Stabilization "
4. Gully control "

1 Retaining Structure

- It is usually a wall constructed for the purpose of supporting vertical or nearly vertical earth bank.
- It is most important in hill road to provide adequate support for road and slope.
- It is constructed on valley side to retain fill materials and also on cut side to prevent ^{landslide} towards road.

Condition where retaining structures are required are:

- (a) Valley side surface get saturated in monsoon.
- (b) Place where under cutting by river or stream.
- (c) In valley point where water flows over the road.
- (d) To achieve road way width where cutting hill is uneconomical.

Shear strength parameter (c, ϕ decrease)

• Empirical design of RW

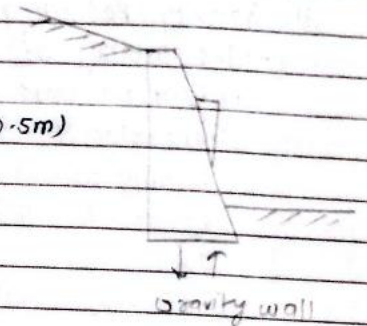
(a) RW upto 6m height

> Base width = $0.5H$ m

> Top width = 0.45 to 0.6 (0.5m)

> Rear side = vertical

> Front side = batter 1 in 4



(b) If $H > 6$ m

> Base width = $(0.4H + 0.3)$ to $(0.5H + 0.6)$ m

The rear face is kept vertical and filled with boulders and stones to improve drainage facility.

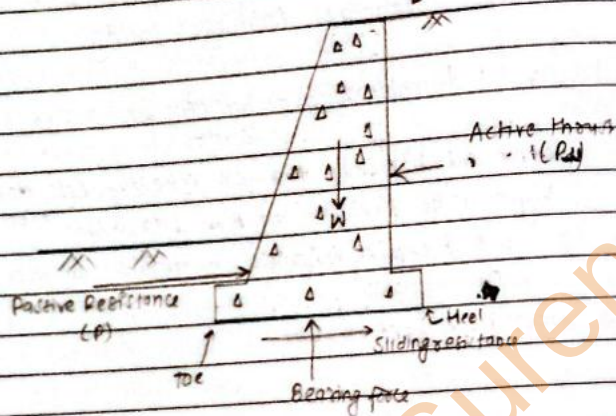
• to avoid tensile crack → backfill in cohesive soil along foot

• water might ^{not} seep through cohesive soil developing hydrostatic pressure on RW (drainage) ^{less permeable}

• so we keep cohesionless material (Gravel, sand) on backfill.

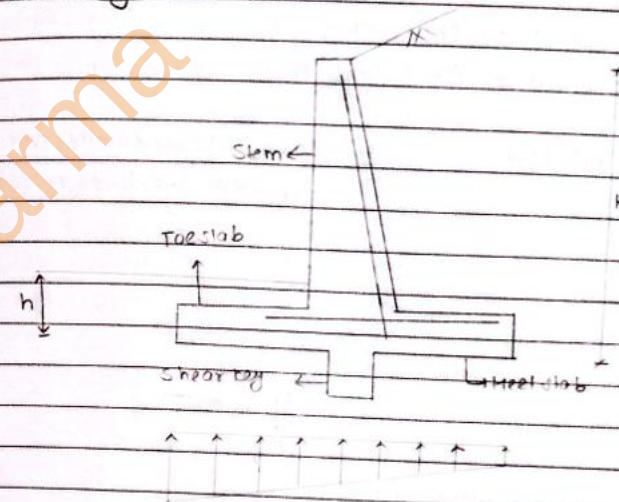
• weep hole kept in RW.

- (i) Gravity Retaining wall
- The stability of this wall is fully depends on its own weight i.e. dead weight
 - Generally, it requires good foundation with sufficient bearing capacity
 - The wall is thicker in bottom and gradually reduced on top
 - It is very cumbersome to construct as it required large amount of materials
 - The size of gravity wall will be reduced by providing reinforcement at tension side.



(ii) Cantilever Retaining Walls

- Has relatively thin stem, usually made of concrete with reinforcement to resist tensile force.
- Reinforcement are provided on both footer and stem side
- Re-bar is placed that footer and stem becomes one integral unit



(iii) Counterfort wall

→ Similar to cantilever wall, but further supported by additional thin triangular shaped walls or counterforts, built at right angles to the stem of wall.

✓ → The counterfort reduces bending moment and shear force in the stem.

→ They are hidden within the earthen or gravel backfill of the wall.

✓ → Economical for height more than 6m.

ve counterfort & backfill same side → counterfort wall

" " " " opposite " → buttress wall.

(iv) Crib wall

✓ → Box type structure built from interlocking struts of timber, precast RCC, steel or timber and usually infilled with soil or stone.

✓ → It is flexible type retaining structure, hence to some extent it can resist differential deformation and settlements.

2

River training structure → guide river flow in definite direction.

Need

(i) To provide safe passage to rain flood w/o overflowing its bank and thus prevent flooding of the adjoining area.

(ii) To prevent the river from changing its course.

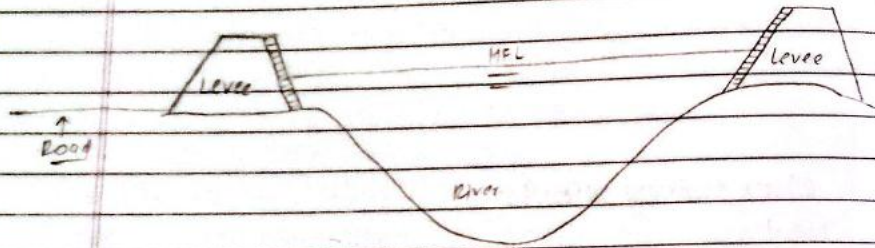
(iii) To prevent erosion of bank and stabilize the river channel.

(iv) To deflect the river flow away from a bank or to attract the river flow toward the bank.

- Methods of River training:
 1. Marginal embankments or levees
 2. Guide banks
 3. Groyne or spur
 4. Artificial cutoff

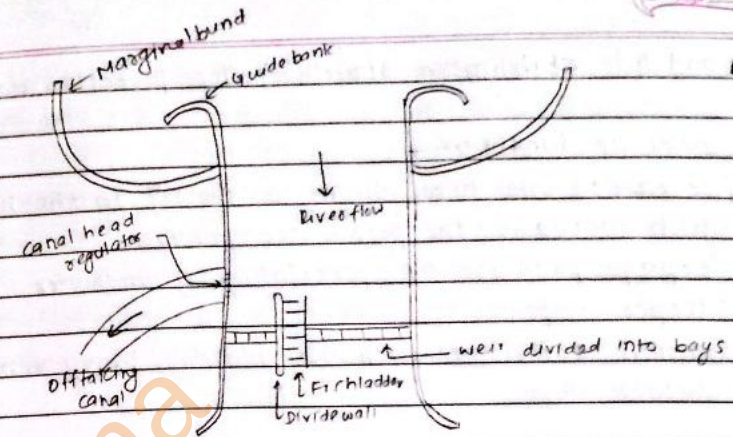
(i) Marginal embankments/levees

- Earthen embankment which are constructed // to the river banks at some suitable distance from it.
- Alignment of levees should follow the normal meandering pattern of river.



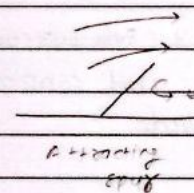
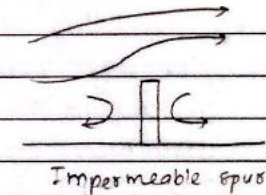
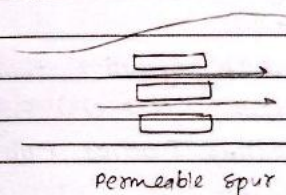
(ii) Guide banks

- Earthen embankments constructed for confining the alluvial river flow within a reasonable waterway and guiding it to a hydraulic structure such as bridge, weir, or barrage, so that there is straight & non tortuous approach to it.

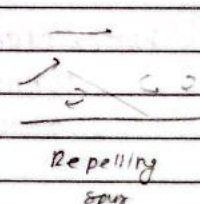
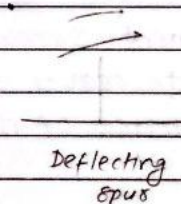


(iii) Spurs

- A spur is a structure made to project flow of river bank into a stream or river with the aim of deflecting the flow away from the side of river on which the groyne is built.



(on opposite bank)



(river repelled to opposite bank)

3. Landslide stabilization structure / Slope protection structure

Causes of landslide:

- Increase in moisture content of the soil in the hill slopes which increases the pore water pressure.
- Seepage pressure of percolating groundwater
- Steeper slope
- Human activities such as blasting, heavy vehicle of unstable slope.

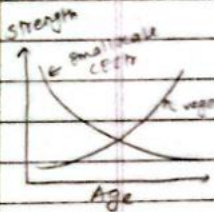
Slope protection work can be categorized into:

1. Control measure
2. Restraint measure
3. Alternative works such as realignment of road

(i) Control measure

(a) Earth work: Cutting and filling must be carried out by providing safe slope.

(b) Bio-engineering: Various types of vegetation structure in conjunction with small scale civil Engineering structures. Eg. Jute netting, Facines, Palisades, Turfing, Brush layering, etc.



(c) Water Management: Surfaces and sub-surface drainage to reduce seepage and consequently reduce chance of slope failure.

(ii) Restraint measure

- a) Slope work: Stone pitching, stone arch formation, dry stone wall.
- (b) Anchoring structure: Rock bolting, soil nailing, etc.
- (c) Retaining structure: Retaining wall, gabion wall, stone masonry, crib wall, etc.
- (d) Protection work: Rock fall wire net, check dam.
- e) Piling work: Steel pile, timber pile, etc.

4 Gully control structures

→ Gullies are highly visible form of soil erosion created by running water.

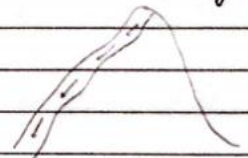
→ They are deep sided water courses, meters to tens of meters in depth and width.

→ Gully formation starts with formation of rills by surface water flowing down a slope especially where soil is exposed.

→ Gullies channel and accelerated runoff contribute to flood and flash flood causing damage to the surrounding area, infrastructure, sediment flow, etc.

The major causes of gully erosion are:

- ✓ > Erosion of catchment
- ✓ > Steep unprotected slopes and drainage channels
- ✓ > Side slope failure due to the cutting of the gully channel.



Gully control structure are:

- (i) Broken rock
- (ii) Bawn log or timber barrier
- (iii) Brick barrier
- (iv) Drop structure
- (v) Chutes

Drainage ~~on~~ works on Hill roads

→ One of the main problem in hill drainage system is the surface water from the hill slope that flows towards the roadway.

→ Road side drains are generally planned and construct for water from roadway surface only.

→ Hence road side drain can not safely discharge the water that comes from hill side slope.

→ Consequently severe damage to the pavement

• Component of surface drainage system in Hill road:

1. Drain of water from hill slope
2. Road side drain
3. Cross drainage structure

(a) # Drainage of water from hill slope

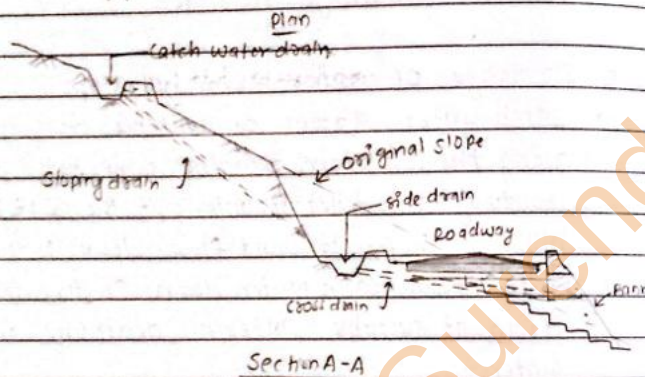
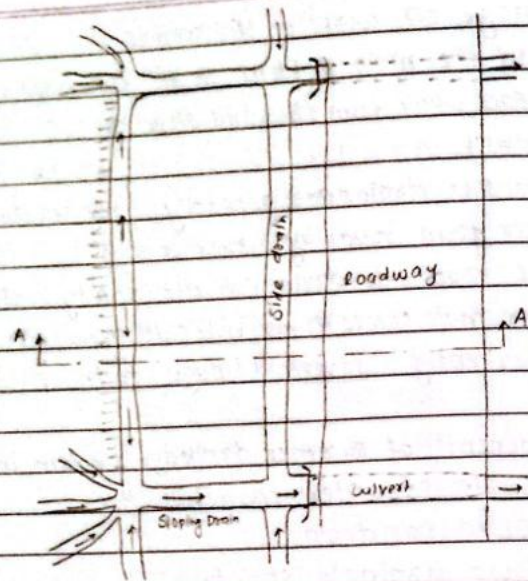
→ Catch water drains of suitable cross section are constructed along the hill slope, running somewhat parallel to the roadway in order to intercept the water flowing down hill slope, before it could flow down to the roadway.

depends
on slope
of hill
& rainfall
amount
→ water
to roof

→ Water from catch water drain is directed by sloping drain at suitable intervals depending on quantity of water.

→ Water from sloping drain is collected in catch pit and finally discharged to water course with the help of cross drain structure such as culvert.

→ The catch drain should be given gradient of $\frac{1}{50}$ to $\frac{1}{33}$ to avoid high velocity.



(b) Road side drain

→ provided only on the hill side of the roads and not on both sides.

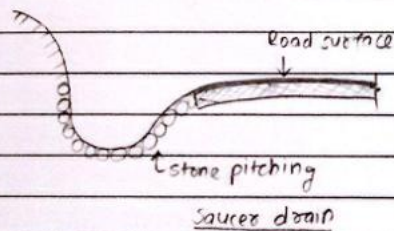
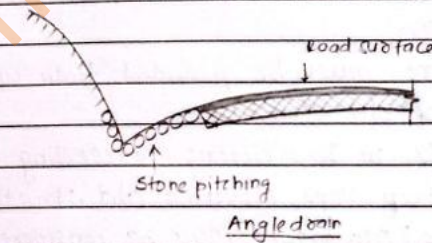
→ Due to limitation in the formation width, the side drains are constructed to such a shape that at emergency the vehicles could utilize the part of the side drain for crossing another vehicles.

→ The usual types of side drains which can serve the purpose as above are:

(a) Angle drain

(b) Saucer drain

(c) kerb and channel drain



(C) Cross Drainage Structures

- As far as possible cross drainage should be taken under the road and at right angle to it.
- At the head of cross drain small catch pits must be provided to collect the stone, soil and to prevent scour.
- The floor of the cross drain is given a longitudinal slope of 1 in 7.
- Waterway required for the cross drainage works may be calculated from usual formula such as Manning's, Chezy, etc.
- As far as possible culvert or single span bridge are constructed.
- Where culvert under the road can not be provided, causeway are alternate solution.
- Protective structure must be provided both upstream and down stream of old structure.
- If the hill road is in box cutting exceeding 5m or road passes very steep slope, elevated old structure called elevated flume/aqueduct may be considered.

①





basic capacity $C = \frac{1000V}{S}$

Design hourly vol. of traffic

traffic lane (Tw)

design vehicle

basic capacity $C = \frac{1000V}{S}$

Design hourly vol. $= \frac{\text{of traffic}}{\text{avg. s vehicle}}$

traffic lane (T_w)

+ design vehicle

basic capacity $C = \frac{1000V}{S}$

Design hourly vol. $= \frac{\text{of traffic}}{\text{avg. s vehicle}}$

traffic lane (T_w)

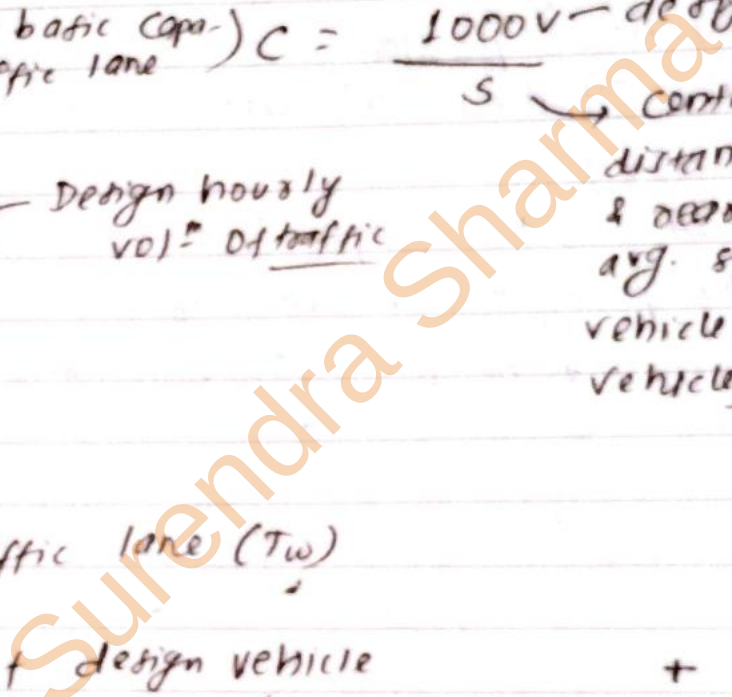
+ design vehicle

basic capacity $C = \frac{1000V}{S}$

Design hourly vol. $= \frac{\text{of traffic}}{\text{avg. s vehicle}}$

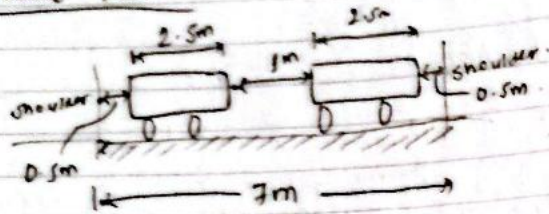
traffic lane (T_w)

+ design vehicle



Single lane $\rightarrow$ 3.25 m per lane

Double lane $\rightarrow$ 8.5 m per lane (for multiple lanes)



Shoulders

- ↳ provided along the road edge
- ↳ emergency parking
- ↳ lateral support to carriage way (pavement)
- ↳ overtaking
- ↳ Increase lateral clearance (thus, inc. visibility)
- ↳ provide space for people to walk
- (Min. 0.75m on each side) $\rightarrow$ NRV 2070

Row

Lay-by

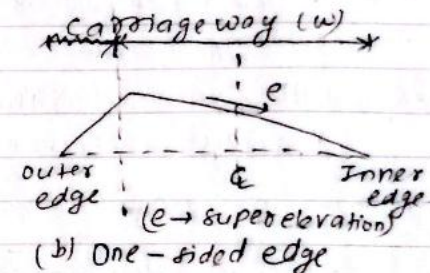
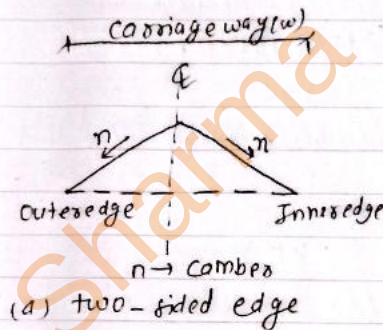


Daily consumptive use of water = 1.1 cm

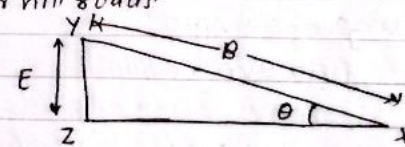
$$\text{Frequency} = \frac{11.9}{1.1} \approx 10 \text{ days}$$

(EP)

Superelevation / Cant / Banking



$\rightarrow (7-10\%) \Rightarrow$ super elevation
for hill roads
or 1.15



$$e = \frac{YZ}{XZ} = \tan \theta$$

$\theta \rightarrow$ very small

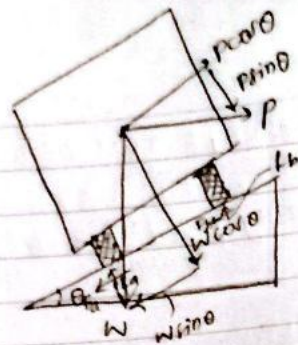
$$e = \tan \theta \approx \sin \theta = \frac{E}{B}$$

• Super elevation rate (e)

- ↳ measured as relative elevation of outer edge of the pavement to width of the pavement.

$\therefore \tan \theta = e$

Analysis of superelevation:



$P \rightarrow$ centrifugal force, kg

$W \rightarrow$ wt. of vehicle, kg

$R, R_g \rightarrow$ Normal rxns of the pavement on the wheels

$F, F_g \rightarrow$ Frictional resistance b/w tire and pavement acting transversely along the pavement surface towards the centre of the curve.

Eqn, Equating all forces $\perp$ to the pavement surface

$$R_p + R_g = W \cos \theta + P \sin \theta$$

Equating all forces $\parallel$ to pavement surface

$$F_g + F_b = P \sin \theta - W \sin \theta$$

$$\text{or, } f(R_p + R_g) = P \cos \theta - W \cos \theta$$

$$W \cos \theta + P \sin \theta$$

$$\text{or, } \frac{P}{W} = \frac{f \cos \theta + \sin \theta}{\cos \theta - f \sin \theta}$$

$$\text{or, } \left[\frac{P}{W} = \frac{f + \tan \theta}{1 - f \tan \theta} = \frac{e + f}{1 - ef} = \frac{V^2}{gR} \right]$$

$$(e = \tan \theta)$$

Since,

For max^m $e = 0.07$, $f = \text{coeff. of lateral friction for bituminous pavement} = 0.15$

Thw, $ef \approx 0$ So, $1 - ef \approx 1$

$V \rightarrow$ design speed (km/h)
 $e \rightarrow$ $\left[\frac{P}{W} = e + f \right] = \frac{V^2}{gR}$ EQUATION OF SUPERELEVATION.

If V in km/h & $g = 9.81 \text{ m/s}^2$

$$[e + f = \frac{V^2}{127R}]$$

Equilibrium superelevation, i.e. for $f = 0$

(I) $[e = \frac{V^2}{127R}]$

↳ pressures on both the wheels of vehicle become equal

↳ centrifugal force is completely balanced by superelevation

↳ vehicle in stable position. So it can be driven at constant steering angle.

(II) # Terrain condition s.t. outer edge cannot be raised i.e. $e = 0$.

$$[f = \frac{V^2}{127R}]$$

{ frictional force only balances centrifugal force }

Speed should be restricted to $[V = \sqrt{127fR}]$

(III) # If $e = 0.07$, $f = 0.15$

Speed should be restricted to

$$\{V_a = \sqrt{127R(e+f)}\}$$

Restricting V for max. $e = 0.07$ min. $f = 0.15$ (bitumen)

Maximum superelevation:-

NR 2070:-

• In plain & rolling terrain $\rightarrow 7\%$

• In snow bound areas $\rightarrow 7\%$

• In hilly areas not bound by snow $\rightarrow 10\%$

$\rightarrow$ higher superelevation provide easy condition for fast moving vehicle but causes inconvenience to slow moving & heavily loaded trucks.

$\left. \begin{array}{l} \text{Min}^m \text{ Superelevation} \rightarrow 0.5\% \\ \text{Min}^m \text{ Gradient} \rightarrow 0.5\% \end{array} \right\} \text{ default}$

NRRS 2071

Road 7%
 Hill 10%

Min^m Superelevation

L to design of surface water
 should be equal to rate of camber of pavement
 (NRRS 2070 & NRRS 2071)

□ Design of Superelevation:

Fast moving $\rightarrow$ higher e
 Slow " $\rightarrow$ lower e .

Design procedure: IRC.

Step 1: Calculate e for 75% of design speed 'V'

with considering coeff. of friction f .

Mixed traffic condition (75% of V)

$$e = \frac{(0.75V)^2}{gR}$$

Step 2: If calculated $e \leq \frac{1}{15}$ (7%) $\rightarrow$ okay

If $e > 7\%$, provide max^m e of 7% and proceed to step 3 and 4.

Step 3: Check coeff. of lateral friction for max^m e .

Step 4: Calc allowable speed for $e = 7\%$ and $f = 0.15$

$$V_a = \sqrt{127 R (e + f)}$$

limited speed (needs warning sign)

Halted max.
$$e = \frac{V^2}{127 R}$$

NURS $\rightarrow e = \frac{V^2}{225 R}$ } i.e. for (0.75V)

(8) The design speed of a highway with mixed traffic is 75 kmph and radius of horizontal curve is 140m. Design superelevation for the highway. If max^m superelevation of 7% is not to be exceeded, calculate the max^m allowable speed on this horizontal curve. Assume coefficient of lateral friction is 0.15.

$\rightarrow$ Solⁿ:

1. Since traffic is mixed, considering reduced speed (0.75V) neglecting f ,

$$e = \frac{(0.75V)^2}{127 R} = 17.8\%$$

2. Since $e_{\text{req}} > 7\%$, provide $e = 7\%$.

3. for $e = 7\%$ & $V = 75$ kmph.

$$e + f = \frac{V^2}{127 R}$$

$$\Rightarrow f = 0.246 > 0.15$$

Thus, 75 kmph is not safe for $e = 7\%$

4. provide $e = 7\%$ and $f = 0.15$

$$V_a = \sqrt{127 R (e + f)} = \frac{62.54 \text{ kmph}}{0.07 \quad 0.15}$$

Recommended speed = 60 kmph

(normally multiple of 5 kmph)
 (provided sign)

Methods of providing superelevation.

* Involves:

- Elimination of crown of the cambered section.
- Rotation of the pavement cross-section to attain full superelevation.

(A) → By rotation of the outer surface.
 ↳ By shifting position of crown (Diagonal crown Method)

(A) → final → to create one way.

(B) → Rotation of roadway surface about its ϕ .
 (rotates with traffic to fill) → balance.

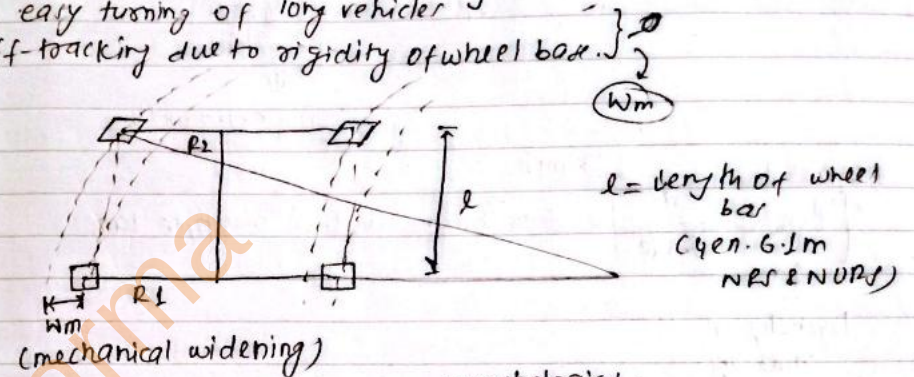
↳ Rotation of roadway surface about its inner edge.
 (a elevation change → discomfort in sight distance,
 (drainage will be best method).
 (cut - fill not balanced)

↳ Rotⁿ of roadway surface about its outer edge.
 (drainage problem)
 (a elevⁿ change)

* Extra Widening

- For safety
- For visibility (outer side)
- For easy turning of long vehicles
- Off-tracking due to rigidity of wheel base.

} Psychological widening



psychological.

Extra widening, $W_e = W_m + W_{psy}$

$$R_2 = R_1 + W_m \quad \text{--- (i)}$$

$$R_2^2 = l^2 + R_1^2 \quad \text{--- (ii)}$$

$$\Rightarrow R_2^2 = l^2 + (R_2 - W_m)^2$$

$$R_2^2 = l^2 + R_2^2 - 2R_2W_m + W_m^2$$

$$\left\{ W_m = \frac{l^2}{(2R_2 - W_m)} \right\}$$

But, $R_1 \approx R_2 \approx R$
 & $W_m \ll R$

So,

$$\left[W_m = \frac{l^2}{2R} \right] \leftarrow \text{required extra widening.}$$

↑ single lane road

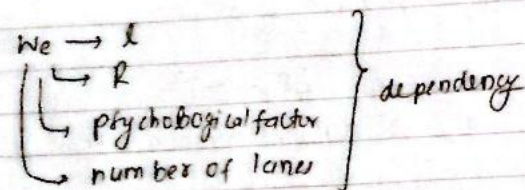
For multiple-lane roads,

$$\left[W_m = \frac{n l^2}{2 R} \right] \rightarrow \begin{array}{c} \underline{n - \text{lanes}} \\ \Downarrow \\ \underline{n - \text{vehicles}} \\ \Downarrow \\ \underline{n - \text{extra width}} \end{array}$$

Empirical $\left[W_{psy} = \frac{V}{9.5 \sqrt{f}} \right] \xleftarrow{\text{kmph}}$ For both single & multiple lane.

↑
Psychological widening.

Thus, $\left[W_e = \frac{m l^2}{2R} + \frac{V}{9.5\sqrt{R}} \right]$



$\therefore W_e \rightarrow$ provided for small radius curve only
(i.e. for $R < 300m$)] N_{20}

$\text{NRRS 2075} \rightarrow \text{only w/m (no Wpsy)}$
 $\downarrow$
 Eval ready.

1) Find the total pavement width required for four lane highway on a horizontal curve of radius 200m. The design speed is 50 kmph and length of the wheel base of vehicle is 6.1m.

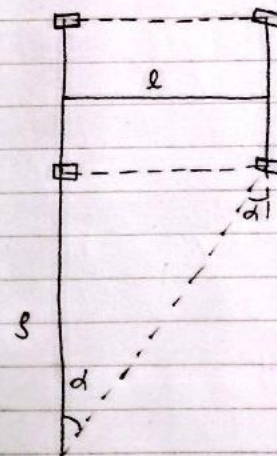
→ Width of carriage-way = $4 \times 3.5 = 14\text{m}$. (As, $R > 300\text{m}$)

$$W_e = \frac{4 \times 6 \text{ J}^2}{2 \times 200} + \frac{50}{9.5 \sqrt{200}} = 0.744 \text{ m}$$

Thus,

total pavement width = $14 + 0.744 = 14.744\text{m}$

Equation of steering curve:
(Why spiral is ideal curve?)



• Assumptions:

- (1) Vehicle is moving with uniform linear speed (v)
- (2) Driver is steering at constant angular velocity (ω)

Translational motion:

$$v = \frac{ds}{dt} \quad [ds = \text{length of translation curve covered in } dt]$$

Rotational motion:

$$\omega = \frac{d\alpha}{dt} \quad (\alpha = \text{steering angle})$$

$$\tan \alpha = \frac{l}{s}$$

Relation of s & l

$$[\because \alpha = \frac{l}{s}] \quad (l \ll s)$$

Taking derivative w.r.t. s ,

$$\frac{d\alpha}{ds} = \frac{-l}{s^2} \quad \text{--- (1)} \Rightarrow d\alpha = \frac{-l}{s^2} ds$$

Taking derivative w.r.t. t ,

$$\frac{d\alpha}{dt} = \frac{-l}{s^2} \frac{ds}{dt} \quad \text{--- (2)}$$

$$\text{or, } \omega = \frac{-l}{s^2} \times \frac{ds}{dt}$$

$$\Rightarrow \omega = -\frac{lv}{s^2} \frac{ds}{ds}$$

$$\text{or, } ds = -\frac{lv}{\omega} \frac{ds}{s^2} \quad \text{--- (3)}$$

Integrating both sides,

$$s = \frac{lv}{\omega} \times \frac{1}{s} + k \quad \text{--- (iii)}$$

Boundary condition, for $s=0$, $l=\infty$, so, $k=0$

Thus,

Length of the transition curve,

$$s = \frac{lv}{\omega} = \frac{C}{s}$$

i.e. $s \propto \frac{1}{s}$ → spiral is ideal for TC.

Length of TC.

(3) For gradual introduction of centrifugal accelⁿ:

l_s be length of TC, t in sec be the time req. to traverse l_s at uniform speed v (m/sec)

$$t = \frac{l_s}{v} \quad \text{--- (5)}$$

$$\text{Max}^m \text{ centrifugal accel}^n = \frac{v^2}{g}$$

$$\text{Rate of change of centrifugal accel}^n = \frac{\frac{v^2}{g} - 0}{t}$$

$$= \frac{\frac{v^2}{g}}{\frac{l_s}{v}}$$

$$= \frac{v^3}{l_s g}$$

If C is the max rate of change of centrifugal accelⁿ within which drivers/passengers do not feel discomfort.

$$[L_s = \frac{V^3}{CR}] \text{ m/s}$$

In NRS: $[L_s = \frac{V^3}{47CR}] \text{ kmph}$

③ → ?

Preferred range:- $[0.5 < C < 0.8 \text{ m/sec}^3]$

• Normally used:

$$[C = \frac{80}{75+V}] \text{ kmph}$$

Empirical value.

But prefer

② For introduction of superelevation

Say, $e \rightarrow$ design superelevation

e be design superelevation, w be extra widening.

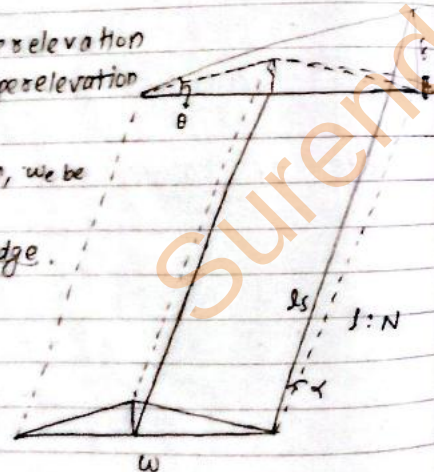
For rotation about inner edge.

$$\tan \theta = e = \frac{E}{W+We} \quad \text{--- (1)}$$

Let, superelevation is to be provided at 1:N ratio,

$$\tan \alpha = \frac{E}{L_s} = \frac{1}{N}$$

$$\text{or, } \left[\frac{e(W+We)}{L_s} = \frac{1}{N} \right]$$



$$[L_s = Ne(W+We)]$$

1:N = 1:150 for plain area

= 1:60 for mountainous & steep area

Limiting values
↑
20 m/sec
30 m/sec
40 m/sec

Rotation about center-line

$$\tan \theta = e = \frac{E}{W+We} \quad \text{--- (1)}$$

$$\tan \alpha = \frac{E/2}{L_s} = \frac{1}{N} \quad \text{--- (2)}$$

$$\Rightarrow [L_s = \frac{Ne(W+We)}{2}]$$

Minimum length required:-

$$[L_s = \frac{2.7V^2}{P}] \text{ kmph}$$

for plain or rolling

$$[L_s = \frac{V^2}{P}] \text{ for mountain or steep}$$

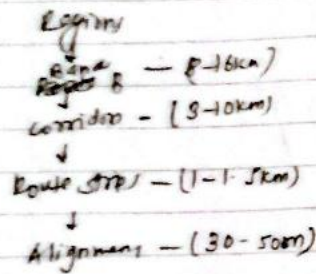
$$\# \underline{L_s = \max \{ (1), (2), (3) \}}$$

Engineering survey for highway Alignment:

- Map study
- Reconnaissance
- Preliminary
- Detailed

② Route location process

↳ Consider the route corridor $\Rightarrow$ route strip



↓
alignment
with minimum
width of ROW

Map study

↳ topographic map study.

↓
Natural +
man-made
features
location identified

3-dimension visualization
(includes contours)

- has obligatory points.
- tentative route can be made.
- ↳ several alternative routes.
- ↳ can be through latest aerial photograph

Scale of topographical map
(in map book or
required area of Nepal)

Scale: 1:25000

Scale: 1:50000

country space

→ can't do detail survey from topographical map.

$$25000 \times 0.005 = 12.5m$$

↓
Scale 1:25000 → on 1:25000

→ At least 3 alternatives marked
for next step reconnaissance survey.

Reconnaissance survey

↳ Map की characteristics में जानें but field में
वर्क की characteristics की study.

↳ simple survey instrument

↳ Approx. gradient, length, curve, No. of drainage, need of
side drain

Need of
hand-pin
board?

↳ Visual soil inspection

↳ Construction materials & labor
(settlement areas near)

↳ Hill slopes, rock types, rainfall, etc.

Modified alignment or verify alignment of previous
alignment from map study

Preliminary survey $\Rightarrow$ Aerial

↳ each alignment on larger scale of survey

↳ more detailed survey.

↳ earthwork quantitative estimation

↳ Best alignment is selected

↓
further study

Q. Road in open-terrain जहाँ पाइंटिंग
done

(Ch. 4)

Highway Drainage
↳ road में enemy → पानी

Transition curve Problem

Q. A two lane pavement highway in hilly terrain has a curve of radius 70m. The design speed is 40kmph.

Determine the

A. length of transition curve

B. total length of curve

C. its tangent length and apex distance.

If the deflection angle is 60 degree. Rate of attainment of superelevation is 1 in 60.

→ Solⁿ:-

$$n = 2$$

hilly terrain

$$R = 70$$

$$S:N = 1:60$$

$$V = 40 \text{ kmph}$$

A. Length of TC:

1. For gradual introduction of C. accelⁿ:-

Rate of change of centrifugal

accelⁿ:

$$C = \frac{80}{75+V} = \frac{80}{75+40}$$

$$[C = 0.696 \text{ m/sec}^3]$$

$$\text{which } 0.5 < C < 0.8 \text{ m/sec}^3$$

$$L_s = \frac{V^3}{47CR} = \frac{40^3}{47 \times 0.696 \times 70} = 28.147 \text{ m.}$$

2. For introduction of superelevation at 1:60.

For design e:

Assuming mixed traffic

condition & 75% design speed,

$$e = \frac{(0.75V)^2}{127R} = \frac{(0.75 \times 40)^2}{127 \times 70} = 10.12\%$$

required $> 7\%$.

So, design provide $ae = 7\%$

Again,

$$f = \frac{V^2}{127R} - e$$

$$= \frac{40^2}{127 \times 70} - 0.07$$

$$= 0.11 < 0.15$$

Thus, $v = 40 \text{ kmph}$ is safe.

extra widening,

$$w_e = \frac{nl^2}{2R} + \frac{V}{9.5\sqrt{R}}$$

length of wheel axle, $l = 6.1 \text{ m}$

$$= \frac{2 \times 6.1^2}{2 \times 70} + \frac{40}{9.5\sqrt{70}}$$

$$= 1.035 \text{ m.}$$

If $V_{\text{limited}} \rightarrow$ Recalculate for ③.

Assuming rot. about inner edge,

$$L_s = Ne(w + e) \quad \text{for lane } 1:N \Rightarrow 1:60$$

$$= 60 \times 0.07 (7 + 1.035)$$

$$= 33.747 \text{ m.}$$

$$\approx 33.75 \text{ m.}$$

3. Minimum length requirement

$$L_s = \frac{V^2}{R} (\text{hilly}) = \frac{40^2}{70} = 22.86 \text{ m.}$$

Thus,

$$\text{Design } L_s = \max. \text{ cases } \{1, 2, 3\}$$

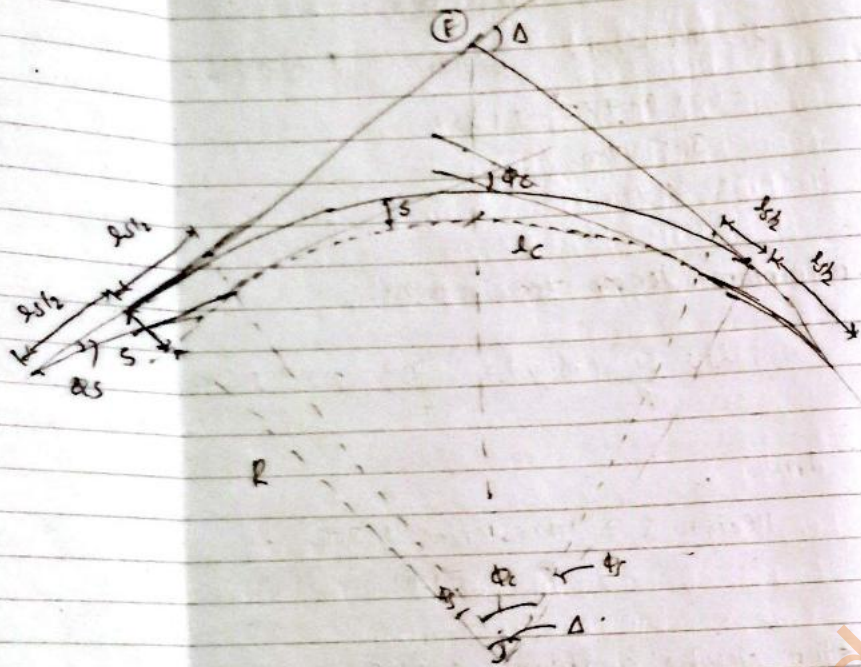
$$= \text{②} = 34 \text{ m.}$$

Transition circular combined curve.

After providing the transition curve, the circular curve is shifted towards inner side by S called lateral shift.

$$\text{Lateral shift } S = \frac{L_s^2}{24R}$$

$R =$ radius of circular curve



$$\text{Spiral angle } (\phi_s) = \frac{l_s}{2R} \times \frac{180^\circ}{\pi}$$

$$\text{Central angle of C. curve } (\phi_c) = \Delta - 2\phi_s$$

$$\text{Length of C. curve } (l_c) = \frac{\pi R \phi_c}{180^\circ}$$

$$\text{Length of combined curve } (l) = 2l_s + l_c$$

$$\text{Tangent length } (T) = (R+s) \tan \frac{\Delta}{2} + \frac{l_s}{2}$$

$$\begin{aligned} \text{Apex distance } (E) &= (R+s) \sec \frac{\Delta}{2} - R \\ &= (R+s) \left[\sec \frac{\Delta}{2} - 1 \right] + s \end{aligned}$$

intd.

B. Total length of curve

$$\text{Spiral angle } (\phi_s) = \frac{l_s}{2R} \times \frac{180^\circ}{\pi}$$

$$= \frac{34}{2 \times 70} \times \frac{180^\circ}{\pi}$$

$$= 13.91^\circ$$

$$\text{Central angle of C. curve } (\phi_c) = \Delta - 2\phi_s$$

$$= 60^\circ - 2 \times 13.91^\circ$$

$$= 32.17^\circ$$

$$\text{Length of C. curve } (l_c) = \frac{\pi R \phi_c}{180}$$

$$= \frac{\pi \times 70 \times 32.17^\circ}{180}$$

$$= 39.304 \text{ m.}$$

$$\text{Length of combined curve } (l) = 2l_s + l_c$$

$$s = \frac{l_s^2}{24l} = 0.69 \text{ m} > 0.25 \text{ m}$$

$$= 2 \times 34 + 39.304$$

$$= 107.304 \text{ m.}$$

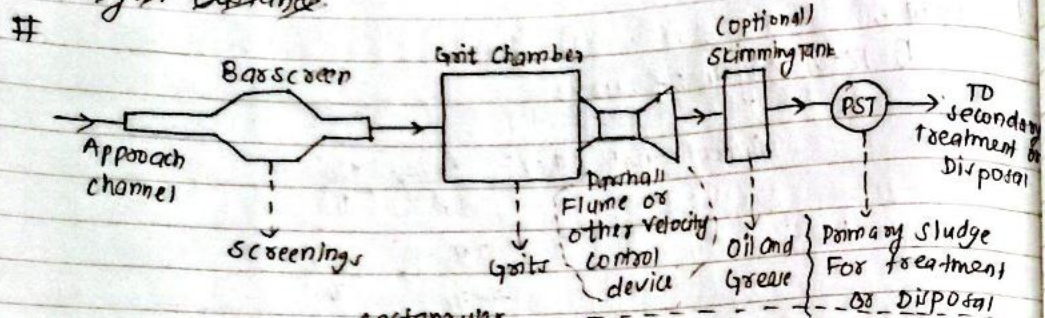
$$\text{C. tangent length } (T) = (R+s) \tan \frac{\Delta}{2} + \frac{l_s}{2}$$

$$= 57.81 \text{ m}$$

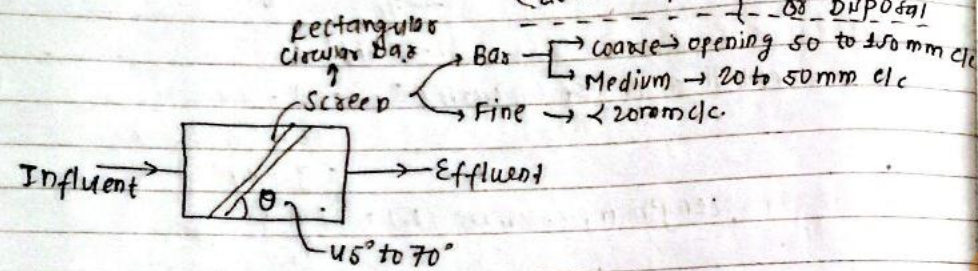
$$\text{Apex distance } (E) = (R+s) \sec \frac{\Delta}{2} - R$$

$$= 11.63 \text{ m}$$

Sight distance

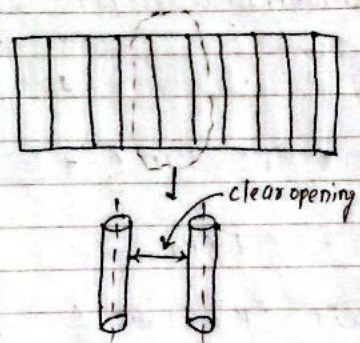


#

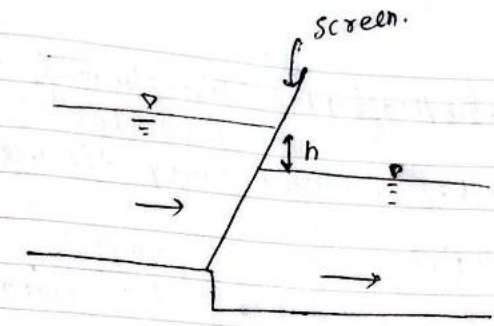


Screening

- ↳ 1st unit WTP/WWTP
- ↳ TO remove floating matter

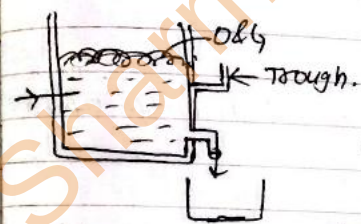


✓ a good option is bar screen followed by fine screen.



Skimming Tank

↳ TO remove O&G, fatty materials, volatile i:-



→ 3 to 15 minute detention time.

→ sometimes, aerated skimming tank may be used.

Analysis of SSD
Sum of reaction/lag distance and braking distance

SSD = Reaction/lag distance + Braking distance

$$\text{Reaction distance (m)} = vt \quad (v \rightarrow \text{m/sec})$$

$$= 0.278vt \quad (v \rightarrow \text{kmph})$$

$v \rightarrow$ speed of vehicle
 $t \rightarrow$ Reaction time (sec)

't' is considered 2.5 sec for the calculation of SSD

Braking distance (l)

Let, 'l' be braking distance, 'f' is coeff. of friction; longitudinal

'W' be weight of the vehicle:

Work done in stopping vehicle = Change in kinetic energy.

$$\text{or, } f \cdot W \cdot l = \frac{1}{2} \left(\frac{W}{g} \right) v^2$$

$$\text{or, } \left[l = \frac{v^2}{2gf} \right] \rightarrow v = \text{m/s}$$

For, $v = \text{kmph}$

$$\left[l = \frac{v^2}{254f} \right]$$

Thus,

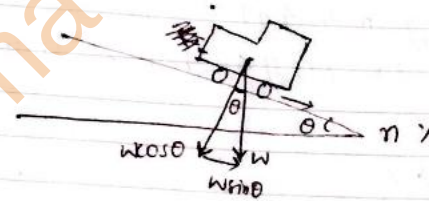
$$\text{Total SSD} = vt + \frac{v^2}{2gf} \text{ m/s}$$

$$= 0.278vt + \frac{v^2}{254f} \text{ kmph.}$$

If η = braking efficiency, $(f \times \eta)$

$$\left[\text{SSD} = 0.278vt + \frac{v^2}{254f\eta} \right] \text{ kmph}$$

$$\left[\text{SSD} = vt + \frac{v^2}{2gf\eta} \right] \text{ m/s}$$



Ascending gradient (माचि जाँहे छ)

Work done in stopping vehicle = Change in K.E.

$$\text{or, } (fW \cos \theta + W \sin \theta) l = \frac{1}{2} \left(\frac{W}{g} \right) v^2$$

For θ very small, $\cos \theta \approx 1$

$\sin \theta \approx \tan \theta (\eta \%)$

Thus,

$$\left[l = \frac{v^2}{2g(f + 0.01\eta)} \right]$$

For descending gradient

$$(fW \cos \theta - W \sin \theta) l = \frac{1}{2} \left(\frac{W}{g} \right) v^2$$

$$\Rightarrow \left[l = \frac{v^2}{2g(f - 0.01\eta)} \right]$$

In general, $SSD = vt + \frac{v^2}{2g(f \pm 0.01n)} \quad \left[v \rightarrow \text{m/s} \right]$

$SSD = \left[0.278Vt + \frac{V^2}{254(f \pm 0.01n)} \right] \quad \left[v \rightarrow \text{kmph} \right]$

Safe stopping sight distance (SSD)

→ For one way or two way movements in same number of lanes

$[SSD = SSD]$

→ For two way movement in opposite direction in single lane

$[SSD = 2SSD] \leftarrow \text{Intermediate sight distance}$

Sight distance P.1

$t = 2.5 \text{ sec}$

$V = 70 \text{ kmph}$

$f = 0.4$

$n = 50\% = 0.5$

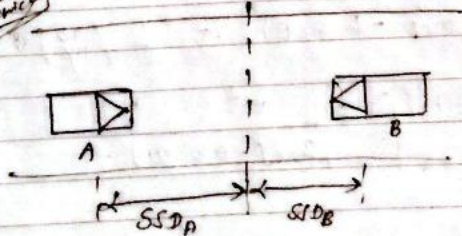
Two buses approaching from opposite direction.

Req. safe sight distance $SSD = 2SSD$

$$= 2 \left[0.278Vt + \frac{V^2}{254(f \pm 0.01n)} \right]$$

2.5 0.4 0.5

231.83 m. 290.24 m.



Overtaking sight distance (OSD)
→ Passing sight distance.

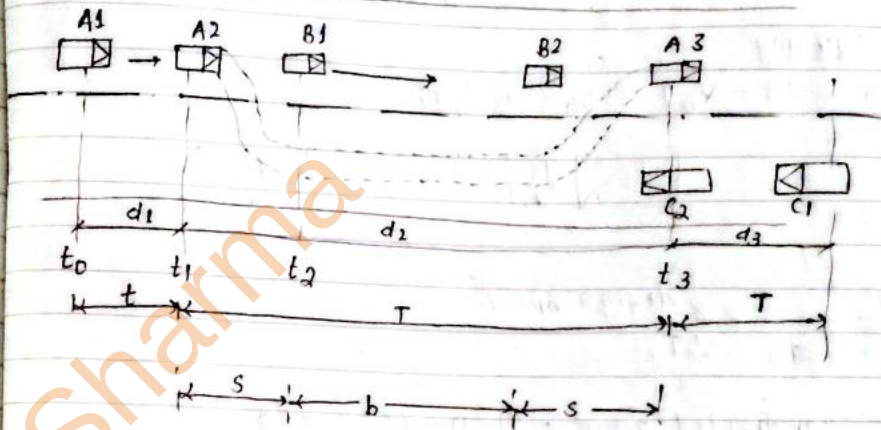


Fig. Analysis of OSD

d_1 :

$d_1 = V_b t \quad \text{--- (1) } \left\{ \begin{array}{l} \text{vehicle A ले भागने speed बताए } \\ \text{'B' को जति प्रवाउद} \end{array} \right\}$

$V_b = V - 16 \text{ kmph}$
 $= V - 4.5 \text{ m/s}$ $V \rightarrow \text{design speed}$

$[t = 2 \text{ sec}] \rightarrow \text{if not given.}$

d_2 :

$d_2 = 2s + b = 2s + V_b * T$

$s = \text{spacing b/w vehicle A \& B before \& after overtaking}$

$= (0.69V_b + 6.1) \text{ m}$

clearance length of vehicle
m/s

$$d_2 = V_b T + \frac{1}{2} a T^2$$

if not given
take $a = 1 \text{ m/s}^2$

uniform overtaking
acceleration. (AASHTO)

Equating.

$$\text{or, } 2s + V_b T = V_b T + \frac{1}{2} a T^2$$

$$\left[T = \sqrt{\frac{4s}{a}} \right] \quad a = \text{overtaking accel.}$$

d_3 : design speed.

$$d_3 = VT$$

Thus, Undivided road: $OSD = d_1 + d_2 + d_3$

Divided road: $OSD = d_1 + d_2$

Q.5 SDP 2

$$V = V_A = 80 \text{ kmph}$$

$$V_b = 65 \text{ kmph}$$

$$a = 3.6 \text{ kmph/sec} = 1 \text{ m/s}^2$$

$$\textcircled{a} \frac{OSD}{2} = \frac{2 \times OSD}{2} = 2 \left[0.278 V_b T + \left(\frac{V_b^2}{2g} \right) \right]$$

$$d_1 = V_b t = 65 \times \frac{2 \times 5}{18} = 36.11 \text{ m.}$$

(Assume $t = 2 \text{ sec}$)

$$d_2 = 2s + V_b T$$

$$s = 0.69 V_b + 6.1 = 0.69 \times 65 \times \frac{5}{18} + 6.1$$

$$= 18.56 \text{ m.}$$

$$T = \sqrt{\frac{4s}{a}} = \sqrt{\frac{4 \times 18.56}{1}} = 8.61 \text{ sec.}$$

$$d_2 = 2 \times 18.56 + 65 \times \frac{5}{18} \times 8.61$$

$$= 192.58 \text{ m.}$$

$$d_3 = VT$$

$$= 80 \times \frac{5}{18} \times 8.61$$

$$= 191.33 \text{ m.}$$

Thus, Required $OSD = d_1 + d_2 + d_3 = 420.02 \text{ m.}$

$\textcircled{b}$ Min length of overtaking zone = 3 OSD
(assuming multiple zone) = 1260.06 m.

Desirable length of oz = 5 OSD
= 2100.1 m.